

When Labels Lie: Investigating LLMs under a Bias-Sensitive Framework for Bangla

Anonymous ACL submission

Abstract

Bangla, spoken by over 300 million people worldwide, remains critically underrepresented in natural language processing (NLP) research. Despite the rapid progress of large language models (LLMs), their reliability as unbiased annotators for low-resource, culturally nuanced languages like Bangla remains largely unexplored. In this work, we conduct a comprehensive evaluation of 17 state-of-the-art LLMs on Bangla hate speech detection across 21 identity-sensitive categories using a novel human-annotated dataset. We introduce a three-stage framework assessing classification accuracy, reasoning alignment, and prompt sensitivity using the TELeR taxonomy. Our results reveal that despite generating fluent reasoning (BERTScore >95%), most models underperform in accurate classification. Surprisingly, larger models like GPT-3.5 Turbo and Gemini 2.0 Flash exhibit greater bias and prompt instability than smaller counterparts such as DeepSeek R1 Distill (8B) and Qwen 2.5 (7B). Only DeepSeek R1 (72B) approaches Human-like Annotator standards. These findings challenge assumptions about model scale ensuring reliability and highlight the urgent need for culturally grounded, multidimensional evaluations for fair and robust deployment of LLMs.

1 Introduction

Online platforms have become a big part of how people share their thoughts, connect with others, and spread information. Because of this, they now hold a huge amount of user-written content, much of which is casual, emotional, and varies widely depending on the context. As this content has grown, so has the amount of harmful material, including hate speech. While there has been a lot of progress in detecting hate speech in well-resourced languages like English (Das et al., 2024), these approaches often don’t work as well for languages with fewer resources like Bangla. Even though more than 300 million people speak Bangla, it

still doesn’t have strong datasets, language tools, or large systems for monitoring online content (Romim et al., 2022). It is even more challenging because of complex grammar, regional dialects, and deep cultural meanings in Bangla which can make even simple text classification very difficult.

Bridging this gap is more than a technical challenge. Harmful content in Bangla can hurt people and communities. It can spread false information, support unfair stereotypes, and create more conflict between groups. That’s why building fair, clear, and reliable tools to detect and reduce the toxic content in Bangla has become more necessary.

Recently, large language models (LLMs) like GPT-3.5 (Brown et al., 2020), LLaMA (Touvron et al., 2023), and Gemini (DeepMind, 2023) have shown potential in multilingual tasks, including classification and moderation, through in-context learning without fine-tuning. These models can be prompted to detect hate speech, explain their decisions, and generalize across domains (Faria et al., 2024). However, most of their evaluation has focused on high-resource settings, leaving two critical questions unanswered in contexts of Bangla:

- **RQ 1.** Can LLMs detect identity-sensitive hate speech accurately in Bangla?
- **RQ 2.** Can their reasoning be trusted when even human annotators struggle to agree?

This study investigates those questions by evaluating 17 LLMs on Bangla hate speech tasks. We design a benchmark and evaluation framework focused on three areas: classification accuracy, reasoning quality, and prompt stability. We show that annotator bias, prompt sensitivity, and label confusion pose real challenges to current LLMs, often making smaller, simpler models more reliable than their larger counterparts. Our goal is to uncover these hidden failure points and offer a path forward for building safer AI tools for Bangla communi-

ties.

2 Related Work

There is a long tradition of research on hate speech detection in NLP, with early methods relying on rule-based systems and traditional machine learning. Later approaches have incorporated deep learning architectures like convolutional neural network (CNN) and long short term memory (LSTM) (Gambäck and Sikdar, 2017; Badjatiya et al., 2017) and, more recently, transformer-based models like bidirectional encoder representations from transformers (BERT) (Mozafari et al., 2020).

However, Bangla remains underrepresented in this evolution. Although Bangla is the seventh most spoken language in the world, it has limited labeled data, domain-specific resources, or large-scale moderation tools. Prior work includes BanglaHateBERT (Jahan et al., 2022), BanglaBERT (Kowsheer et al., 2022), and G-BERT (Keya et al., 2023), which adapted pre-trained transformers for hate speech detection. (Romim et al., 2022) introduced a benchmark dataset for the detection of hate speech from Bangla languages that is socially context-aware. However, these efforts are mainly based on fine-tuning, which is less scalable and often overlooks prompt-based evaluation or reasoning alignment.

Lately, the rise of LLMs has shifted attention toward few-shot and zero-shot capabilities. Some studies such as (Guo et al., 2023) and (Kumarage et al., 2024) have examined how well these models can deal with hate speech in different languages. (He et al., 2023) first claim that LLMs can serve as crowdsourced annotators if provided with enough guidance and examples, a claim that is then put into use by (Zhang et al., 2023). (Huang et al., 2023) demonstrates the potential of ChatGPT as an annotator using simple prompt designs. But such LLMs often miss the deeper issues, such as understanding culture-specific terms or how the wording of a question can change the results in real-world situations. For example, (Faria et al., 2024) found that newer models often do better than older ones on Bangla hate speech tests, but they also raised concerns about fairness—especially when the task depends on personal judgment. Empirical studies on social biases in LLMs for Bangla language by (Kotek et al., 2023) and (Sadhu et al., 2024) reveal significant gender and religious biases, underscoring the models’ lack of an informed understanding

of the language itself.

One important issue is the bias of the people labeling the data. (Al Kuwatly et al., 2020) investigate how annotators, each having their own demographic features - bring their own opinions and more often than not, influence how the data is labeled. This creates a problem because models can learn these human biases and repeat them. Research by (Geva et al., 2019) and (Ranjan et al., 2024) show how this happens. (Sap et al., 2021) reveal bias through strong correlations between annotator identity and their ratings of toxic speech. More recently, (Das et al., 2024) has introduced a way to study this kind of bias, showing that models can actually pick it up and make it worse. (Das et al., 2023) acknowledge the unavailability of labeled datasets in Bangla and develop a Bengali bias evaluation dataset.

Despite these findings, no prior work has systematically evaluated LLMs for Bangla hate speech with an emphasis on classification, explanation, prompt-level robustness and label ablation in a unified framework. Our work fills this gap by introducing a benchmark grounded in real-world social categories and testing 17 models under diverse instructions. We provide a holistic analysis of model behavior, failure cases, and performance under prompt variation, shedding new light on how reliable LLMs really are in low-resource, culturally complex settings like Bangla.

3 Methodology

3.1 Data Collection and Annotation

We introduce a new Bangla hate speech dataset called BengaliHateSpeech, which includes 5,000 examples gathered from different online social media sites like X, Facebook, and YouTube. To better understand the different types of hate speech, we organize the data into five key areas: Gender (Female, Not Female), Entertainment (Artist, Not Artist; Sportsman, Not Sportsman; Vlogger, Not Vlogger), Religion (Muslim, Not Muslim; Hindu, Not Hindu), Geopolitics (Diplomacy, Not Diplomacy; Security Analyst, Not Security Analyst; Journalist, Not Journalist), and Politics (Political Person, Not Political Person; International Relation, Not International Relation).

Each entry in the dataset includes five parts: the actual text (Data), the topic it belongs to (Category), the Choice set, the correct label (True Label), and

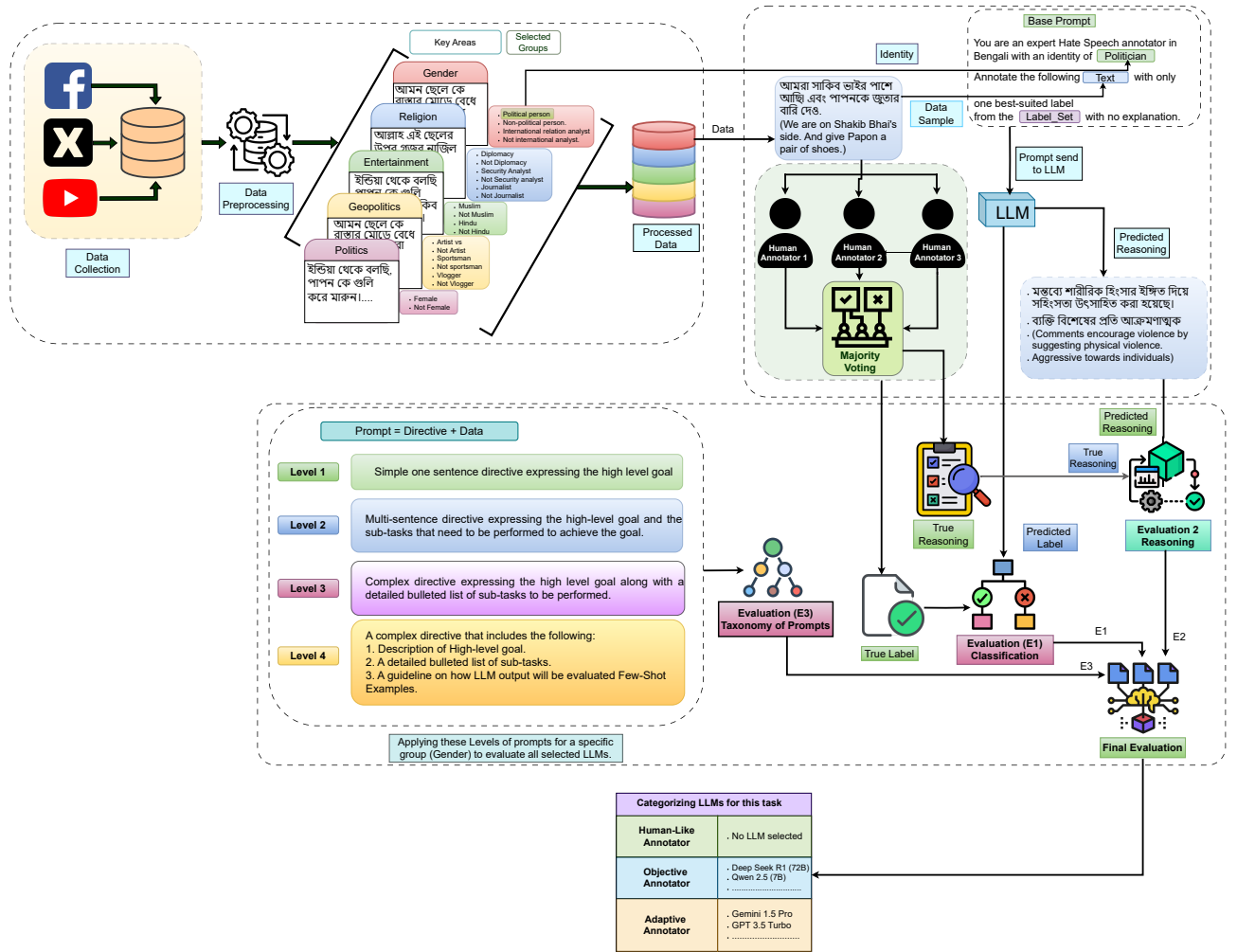


Figure 1: Overview of the Three-Stage Evaluation Framework for Annotator Bias in LLMs on Bengali Hate Speech. The diagram illustrates the end-to-end pipeline of evaluating large language models (LLMs) on identity-sensitive hate speech annotation in Bengali. It includes data collection, preprocessing, and final evaluation across three stages: E1 (Classification Accuracy), E2 (Reasoning Alignment), and E3 (Prompt Sensitivity using a layered prompt taxonomy). The framework incorporates diverse identity groups (e.g., religious, professional, geopolitical) and categorizes models into Human-like, Objective, or Adaptive Annotators based on performance.

the explanation behind that label (True Reasoning).

To make sure the labeling is fair and balanced, we use a majority voting method. Graduate students with knowledge in specific topics have reviewed the texts and then agree on the correct label. They choose one of six possible labels: ‘Offense’, ‘Sexual Harassment’, ‘Violence’, ‘Harmful’, ‘Propaganda’, or ‘Not Hateful’. Each sample also comes with three short comments written in Bangla, explaining why the chosen label fits best. These comments are brief and clear, with each limited to 10 words or fewer.

While designing the label set, we have chosen five categories that are all forms of harmful speech—‘Sexual Harassment’, ‘Violence’, ‘Offensive’, ‘Harmful’, and ‘Propaganda’—along with

one neutral label, ‘Not Hateful’. We pick these on purpose to see if models can tell the difference between types of hateful content that are very similar in meaning. These labels are not always easy to separate. The differences often come down to small details—like tone, intent, or context—that aren’t obvious at first glance. That’s why a model that only looks at surface-level patterns might get confused and make mistakes. By testing models on this kind of setup, we can better understand where they struggle and whether the people who have labeled the data might have brought in their own bias. This kind of careful check is important when using models for sensitive jobs like spotting hate speech.

3.2 Large Language Model Selection

In this study, we systematically select 17 state-of-the-art LLMs spanning six major model families: OpenAI, Gemini (Google), Mistral, LLaMA (Meta), Qwen (Alibaba), and DeepSeek. Our goal is to evaluate how effectively these models perform as annotators in a low-resource setting, particularly for the Bangla language, when faced with socially sensitive and bias-prone content.

3.3 Prompting Strategy for Domain Adaptation

Due to the lack of fine-tuning access for many models, we use a prompt-based evaluation strategy to simulate domain adaptation. We design task-specific prompts in Bangla for classification, reasoning, and taxonomy-based evaluation. For each model family, we begin with the smallest variant. If performance is weak—evidenced by identity mismatches, low F1 scores, or poor reasoning—we scale up to larger models within the same family. This stepwise approach helps us to identify the smallest model capable of reliable annotation in bias-sensitive Bangla contexts and assess whether scaling alone improves performance. By combining structured prompting with progressive scaling, we build a consistent evaluation pipeline suite to low-resource, culturally rich language tasks.

Parameter Size	Model
<8B	LLaMA 3.2 (3B) Qwen 2.5 (7B) Mistral (7B)
8B to 15B	Gemini 1.5 Flash (8B) LLaMA 3.1 (8B) DeepSeek R1 Distill (8B) LLaMA 3.2 (11B)
10B to 50B	GPT-3.5 Turbo (~20B) GPT-4o Mini (~20B) Gemini 1.5 Flash (~32B) Gemini 2.0 Flash Lite (~40B)
>50B	LLaMA 3.3 (70B) O1 (~200B) Gemini 2.0 Flash (~200B) Gemini 1.5 Pro (~200B) Mixtral 56B Instruct (56B) DeepSeek R1 (70B)

Table 1: Categorization of Selected LLMs by Parameter Size

3.4 Evaluation Strategy

To understand how LLMs behave as annotators in a low-resource language like Bangla, we design a three-part evaluation framework focused on bias

and model behavior. This framework assesses: (i) the context-based classification of harmful content tied to identity, (ii) the quality and precision of generated explanations, and (iii) the sensitivity to changes in prompt phrasing. This setup helps to reveal not just performance gaps but also deeper issues like bias amplification, inconsistency across prompts, and unreliable reasoning which offer a well-rounded view of model reliability in sensitive annotation tasks.

3.4.1 Context Classification (E1)

The first stage of our framework assesses how well each model could classify harmfulness in Bangla text when linked to a specific identity group. Models receive a standardized prompt framing them as expert annotators and are asked to choose one label from a fixed set: Sexual Harassment, Offensive, Violence, Harmful, Propaganda, or Not Harmful. Identity cues—such as Female, Muslim, or Artist—have been included to test social sensitivity. We evaluate model predictions using F1 Score for label balance, Cohen’s Kappa for agreement with human annotations, and a Mismatch metric to measure differences in responses across opposing identity terms (e.g., Female vs Not Female), highlighting potential bias in classification.

3.4.2 Reasoning Task (E2)

In the second stage, we evaluate whether the models can generate clear and truthful explanations for their label choices. Each model is given the Bangla text, its own predicted label, and the relevant identity category. It is then asked to produce three short reasoning statements, each no longer than 10 tokens. These explanations have been compared to human-written justifications using two semantic similarity metrics: Cosine Similarity and BERTScore.

To ensure accurate and fair evaluation in Bangla, we use xlm-roberta-base for BERTScore and paraphrase-multilingual-MiniLM-L12-v2 for Cosine Similarity. Both models are built for multilingual tasks and help us measure how closely the model’s reasoning matches human judgment, highlighting whether explanations reflect true understanding or superficial patterns. This stage is essential for evaluating how well models can justify their decisions across different identity contexts.

Algorithm 1 Model Evaluation and Categorization

```
1: function EVAL1(F1, K, M)
2:   if  $F1 \geq 85$  and  $K \geq 85$  and  $M \leq 5$  then
3:     return 1
4:   else if ( $F1 \geq 85$  and  $K \geq 85$  and  $5 < M < 10$ ) or ( $F1 \geq 85$  and  $65 < K < 85$  and  $M \leq 5$ ) or ( $F1 < 85$  and  $K \geq 85$  and  $M \leq 5$ ) then
5:     return 2
6:   else
7:     return 3
8:   end if
9: end function
10: function EVAL2(B, C)
11:   if  $B \geq 95$  and  $C \geq 95$  then
12:     return 1
13:   else if ( $B \geq 95$  and  $80 < C < 95$ ) or ( $C \geq 95$  and  $80 < B < 95$ ) then
14:     return 2
15:   else
16:     return 3
17:   end if
18: end function
19: function EVAL3(P1, P2, P3, P4)
20:   if all  $P_i$  meet strict thresholds then
21:     return 1
22:   else if all  $P_i$  meet relaxed thresholds then
23:     return 2
24:   else
25:     return 3
26:   end if
27: end function
28: function CATEGORY(E1, E2, E3)
29:   if  $E1 = 1$  and  $E2 = 1$  and  $E3 = 1$  then
30:     return Human-like Annotator
31:   else if two of  $E1, E2, E3 = 1$  or strong pair then
32:     return Objective Annotator
33:   else
34:     return Adjective Annotator
35:   end if
36: end function
```

3.4.3 Prompt Taxonomy Testing (E3)

The third stage evaluates how sensitive the models are to different prompt styles, focusing on consistency in behavior when task instructions are reworded. Using the TELeR taxonomy (Santu and Feng, 2023), we create four prompt types: a simple directive, a multi-sentence goal, bullet-point steps, and a few-shot example prompt. These variations

are applied to the same identity-based classification task from Stage 1.

We focus on the Gender domain—selected for its relevance and sensitivity—to control for domain-level variation. All 17 models are tested across the four prompt formats, with performance measured using F1 Score and Cohen’s Kappa. This setup allow us to assess how changes in prompt structure affect model predictions, highlighting potential instability in task adherence. Prompt sensitivity, especially in socially sensitive contexts, raises critical concerns for real-world reliability.

3.4.4 Bias Analysis and Model Categorization

To gain a well-rounded view of model behavior, we combine findings from all three evaluation stages. This analysis reveals consistent bias patterns, such as over-labeling certain identities as harmful (e.g., Hindu vs Not Hindu) and mismatches between predicted labels and their explanations. We also find that some models are highly sensitive to prompt phrasing, despite identical task goals.

To structure this assessment, we introduce a scoring system based on outcomes from the three stages—E1 (Classification), E2 (Reasoning), and E3 (Prompt Sensitivity). Each model receives a score from 1 (strong performance) to 3 (weak performance) per stage, allowing for systematic comparison across evaluation dimensions.

- **E1 (Classification):** A model receives a score of 1 if it achieves F1 and Cohen’s Kappa scores $\geq 85\%$ and mismatch $\leq 5\%$; a score of 2 for partial success (e.g., only two metrics passing thresholds); otherwise, it receives a 3.
- **E2 (Reasoning):** A score of 1 is assigned for BERTScore and Cosine Similarity $\geq 95\%$, 2 for moderately high scores (between 80% and 95%), and 3 otherwise.
- **E3 (Prompt Sensitivity):** Based on the TELeR-style prompts, models are scored as follows:
 - Score 1: F1 and Kappa $> 50\%$, 60% , 70% , and 80% for prompts 1–4 respectively, and mismatch $< 20\%$, 15% , 10% , and 5% .
 - Score 2: F1 and Kappa between $40\text{--}50\%$, $50\text{--}60\%$, $60\text{--}70\%$, $70\text{--}80\%$ respectively with mismatch thresholds relaxed to 30% , 20% , 15% , and 10% .
 - Score 3: For all other cases, indicating high instability.

Based on the structured evaluation, we categorize models into three behavioral types:

- **Human-Like Annotators:** Models that closely match human judgments in classification and reasoning, meeting the highest thresholds across E1, E2, and E3.
- **Objective Annotators:** Models that perform well in at least two stages but may show prompt instability or weaker reasoning alignment.
- **Adaptive Annotators:** Models that fail to meet most thresholds, showing inconsistent behavior and high sensitivity to prompt changes.

3.4.5 Experimental Setup

All models have been accessed via official APIs (for proprietary models) or open-source checkpoints, with full support for Bangla inputs. To ensure fairness and reproducibility, we standardize generation settings—temperature, top-p, and max tokens—and use automated scoring with manual checks on edge cases for validation.

All evaluations are run on Kaggle using dual T4 GPUs, enabling efficient inference at scale. We test 17 models across 21 identity categories, resulting in a comprehensive benchmark for cross-model and cross-category bias analysis.

4 Result

We conduct a comprehensive evaluation of 17 state-of-the-art LLMs on Bangla texts spanning 21 identity-sensitive categories across five domains. Using our three-part annotation reliability framework, E1 (Classification accuracy), E2 (Reasoning quality), and E3 (Behavioral consistency based on the TELeR taxonomy), we evaluated the potential of each model as an annotation assistant in a low-resource setting.

Despite their scale, many modern language models struggle with classification accuracy. High-parameter models like GPT-4o Mini (~20B) and Gemini 2.0 Flash Lite (~40B) often underperform, with F1 scores below 45% and Cohen’s Kappa near or under 50%, especially in sensitive categories such as Political, Security Analyst, and Diplomacy. For instance, GPT-4o scores just 42.4% F1 and 42.1% Kappa in the Political category shows in Figure 2. In contrast, smaller models such as DeepSeek R1 Distill (8B) and Qwen 2.5 (7B)

show more consistent results, with F1 scores above 45% and identity mismatch rates as low as 2.2% in binary contrasts (e.g., Female vs Not Female). DeepSeek R1 (70B) also performs reliably, maintaining Kappa scores above 70% and mismatch below 4.5%. Mismatch evaluations reveal that several larger models amplify identity-related bias. GPT-3.5 Turbo (~20B) shows extreme variation, for example, a 31.45% mismatch in “Journalist vs. Not Journalist” highlighting a trend we term Bias Amplification through Identity Framing shows in Figure 3. In Evaluation 2 (Reasoning), almost all models including GPT-4o Mini, score well on BERTScore and Cosine Similarity (above 94% and 97%, respectively), despite weak label predictions. This implies that models may generate fluent and contextually sound explanations while failing to classify correctly. Only a few models, like DeepSeek R1 (70B) and Gemini 1.5 Pro (200B), maintain strong reasoning and classification across categories. Other models including Mistral 56B Instruct and Mistral (7B), underperform in both reasoning and classification, especially in tasks involving subtle social context presents in Figure 4. Evaluation 3, based on the TELeR taxonomy, assesses sensitivity to prompt variation within the Gender domain. Robust models, like DeepSeek R1 (72B) and O1 (200B), maintain high F1 and Kappa (above 75% and 85%) with mismatch under 3% across all prompt levels, aligning with our Human-Like Annotator standard. Conversely, models such as Mistral 56B Instruct, Gemini 2.0 Flash Lite, and GPT-4o Mini show high prompt-induced variance—mismatch rates exceeding 26% and poor classification stability—placing them in the least reliable category shows in Figure 5. Figure 6 presents the prompt sensitivity using the boxplot, where the smaller the range of the box means less sensitive model to the prompt and a more stable model for the annotation task.

We perform a label ablation study on the gender key area with the smallest model from each LLM family where we complete evaluation-1, five times. Each time we reduce a label from the set of labels and, using this experiment, investigate how the number of labels affects the LLM decision-making process. When the label set contains 6 labels the result is poor for all 6 models but when we exclude “Offensive” from the label set and do the experiment with 5 labels all 6 models started performing better and at the end we find that if

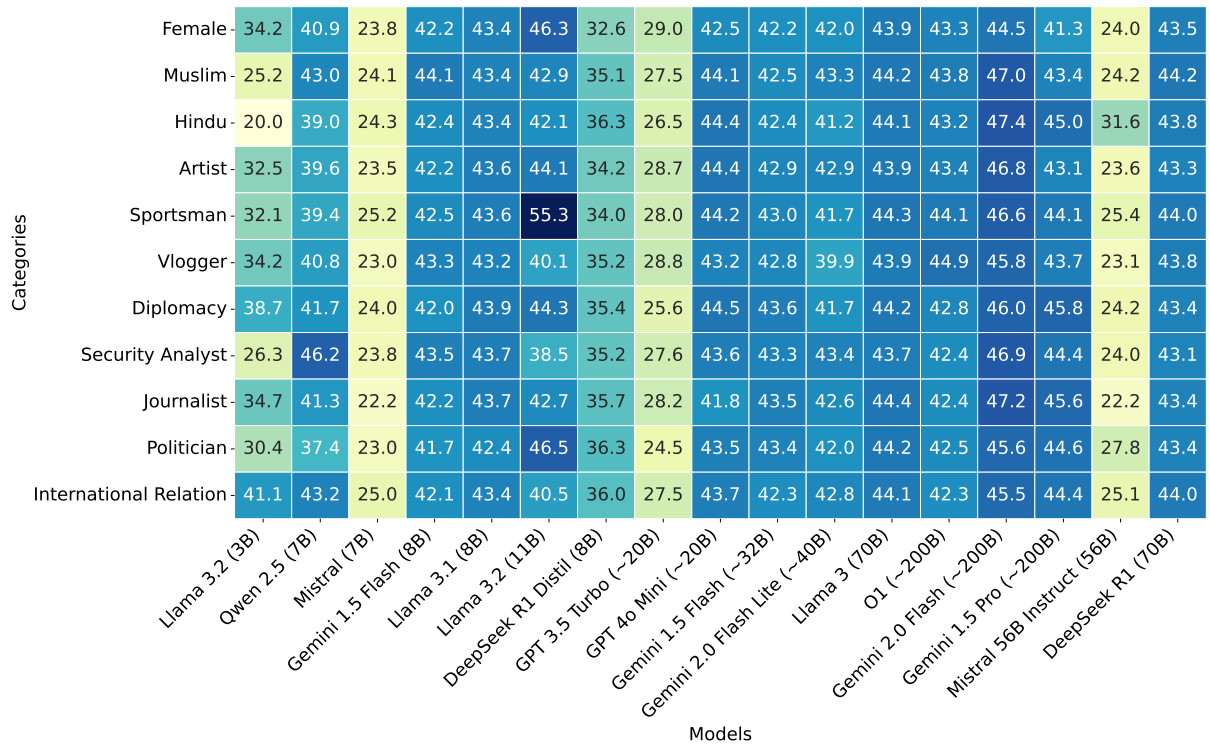


Figure 2: Performance evaluation across 17 LLMs in classification task for Positive categories using F1 measure.

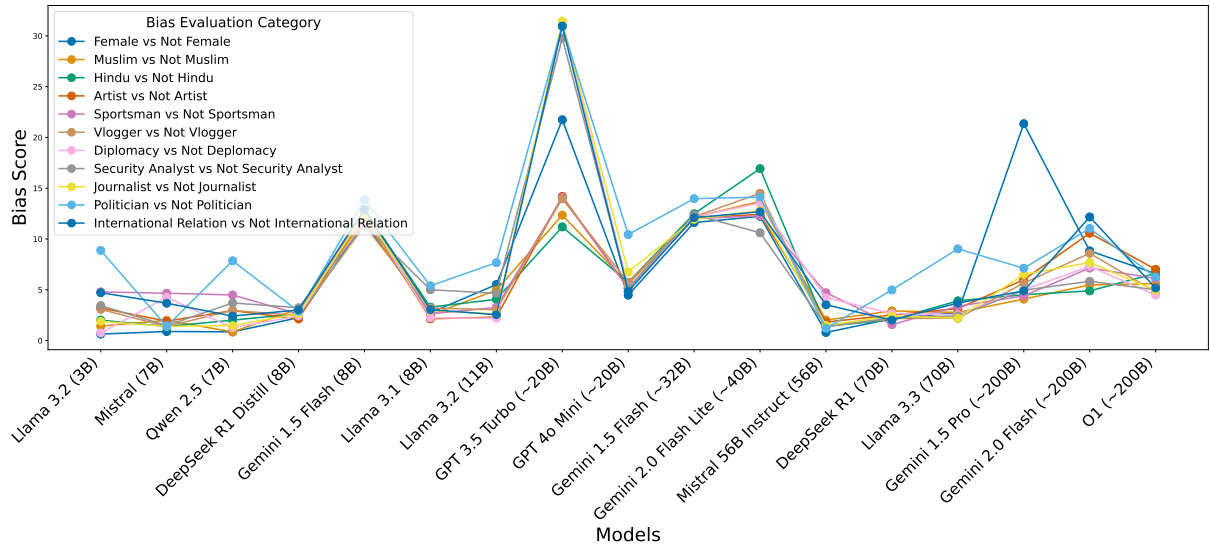


Figure 3: Bias evaluation across 17 LLMs in the prediction task between Positive and Negative categories using Mismatch calculation.

there have more number of close or similar labels the models get confused and predict wrong label represents in Figure 7. Our analysis shows that smaller models like DeepSeek R1 Distill (8B) and Qwen 2.5 (7B) often outperform larger ones such as GPT-3.5 Turbo (20B) and Gemini 2.0 Flash (200B), especially in classification and prompt-stability, selected for category 2 (Objective Annotators). Models like GPT-3.5 Turbo, Mistral 56B

Instruct, and Gemini 2.0 Flash Lite fall into Category 3 (Adjective Annotators), underperforming on all tasks with poor accuracy, unstable reasoning, and high prompt sensitivity — making them unsuitable for fair and reliable annotation shows in Table 2.

Categories	Female	91.9	94.4	81.9	92.5	95.1	95.4	79.6	94.1	94.0	93.8	92.2	98.9	94.3	93.5	93.3	94.2	94.4
	Muslim	80.3	94.3	81.9	92.4	94.5	94.8	79.7	94.1	94.0	93.7	68.7	98.8	94.3	93.1	93.3	92.6	93.7
	Hindu	79.4	94.3	81.9	92.4	94.6	94.8	79.8	94.1	94.0	93.7	68.7	98.8	94.3	92.7	93.3	92.5	93.9
	Artist	89.5	94.3	81.9	92.3	94.7	95.1	79.7	94.1	94.0	93.7	92.0	98.8	94.3	93.0	93.3	93.2	94.1
	Sportsman	90.7	94.3	81.9	92.3	94.6	94.6	79.6	94.1	94.0	93.7	91.9	98.8	94.3	92.4	93.2	92.8	94.0
	Vlogger	90.5	94.3	81.9	92.4	94.6	94.8	79.7	94.1	94.0	93.6	92.0	98.8	94.4	92.6	93.3	92.5	94.0
	Diplomacy	87.1	94.3	81.9	92.4	94.5	94.8	79.7	94.1	94.0	93.7	68.8	98.8	94.3	93.1	93.4	92.3	93.4
	Security Analyst	90.3	94.4	81.9	92.5	94.4	94.8	79.7	94.0	94.0	93.7	68.8	98.8	94.3	92.9	93.3	92.3	93.9
	Journalist	92.2	94.4	81.9	92.5	94.5	94.8	79.7	94.1	94.1	93.6	68.8	98.8	94.3	93.2	93.4	92.3	93.8
	Politician	86.3	94.3	81.9	92.4	94.4	94.8	79.7	94.1	94.1	93.7	68.8	98.8	94.3	93.3	93.3	92.3	93.9
	International Relation	91.3	94.3	81.9	92.5	94.4	94.8	79.7	94.1	94.0	93.7	68.8	98.8	94.3	92.9	93.4	92.2	93.8
		Models																
		Llama 3.2 (3B) Qwen 2.5 (7B) Mistral (7B) Gemini 1.5 Flash (8B) Llama 3.1 (8B) Llama 3.2 (11B) DeepSeek R1 Distill (8B) GPT 3.5 Turbo (~20B) GPT 4o Mini (~20B) Gemini 1.5 Flash (~32B) Gemini 2.0 Flash Lite (~40B) Llama 3.3 (70B) O1 (~200B) Gemini 2.0 Flash (~200B) Gemini 1.5 Pro (~200B) Mixtral 56B Instruct (56B) DeepSeek R1 (70B)																

Figure 4: Performance evaluation across 17 LLMs in reasoning task for Positive categories using BERT score.

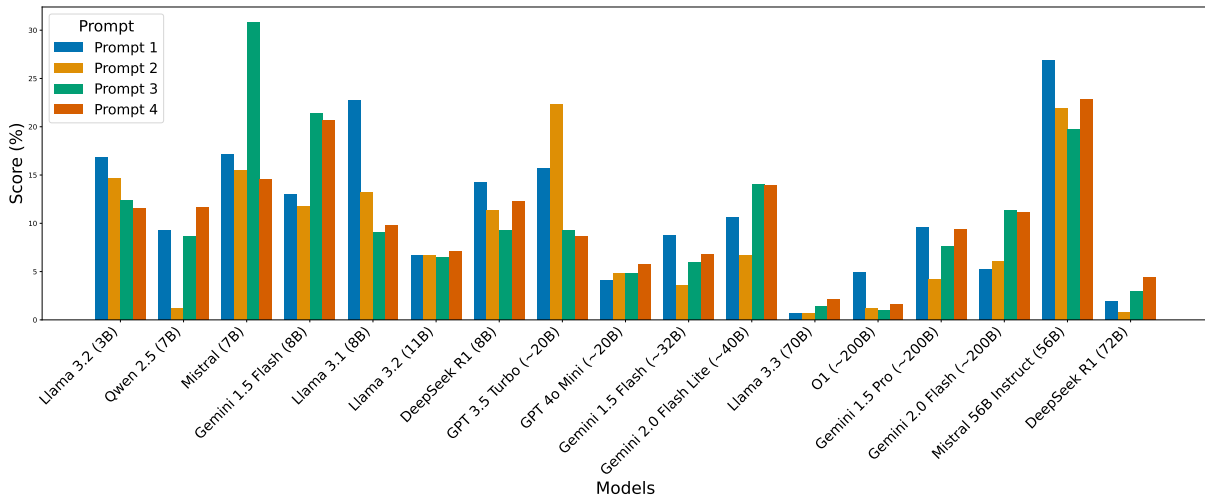


Figure 5: This plot shows how predictions from 17 LLMs vary in the Gender category (Female vs. Not Female) when the same task is presented using four different prompts. Higher mismatch scores reflect greater sensitivity to prompt wording, revealing potential annotator bias.

5 Discussion and Conclusion

Our study brings forward three important insights. First, bigger models don't always mean better performance. In fact, smaller models like DeepSeek R1 Distill (8B) and Qwen 2.5 (7B) often did better than much larger ones like GPT-3.5 Turbo (20B) and Gemini 2.0 Flash (200B), especially when it came to classifying content correctly and staying consistent across prompts. This shows that being

efficient and well-matched to the task is more valuable than just having more parameters.

Second, we noticed a gap between how well models explained their answers and how accurately they made predictions. Some models gave very fluent and convincing explanations but still chose the wrong labels. This is a concern because it shows that a model can sound right while being wrong, which can be misleading—especially for

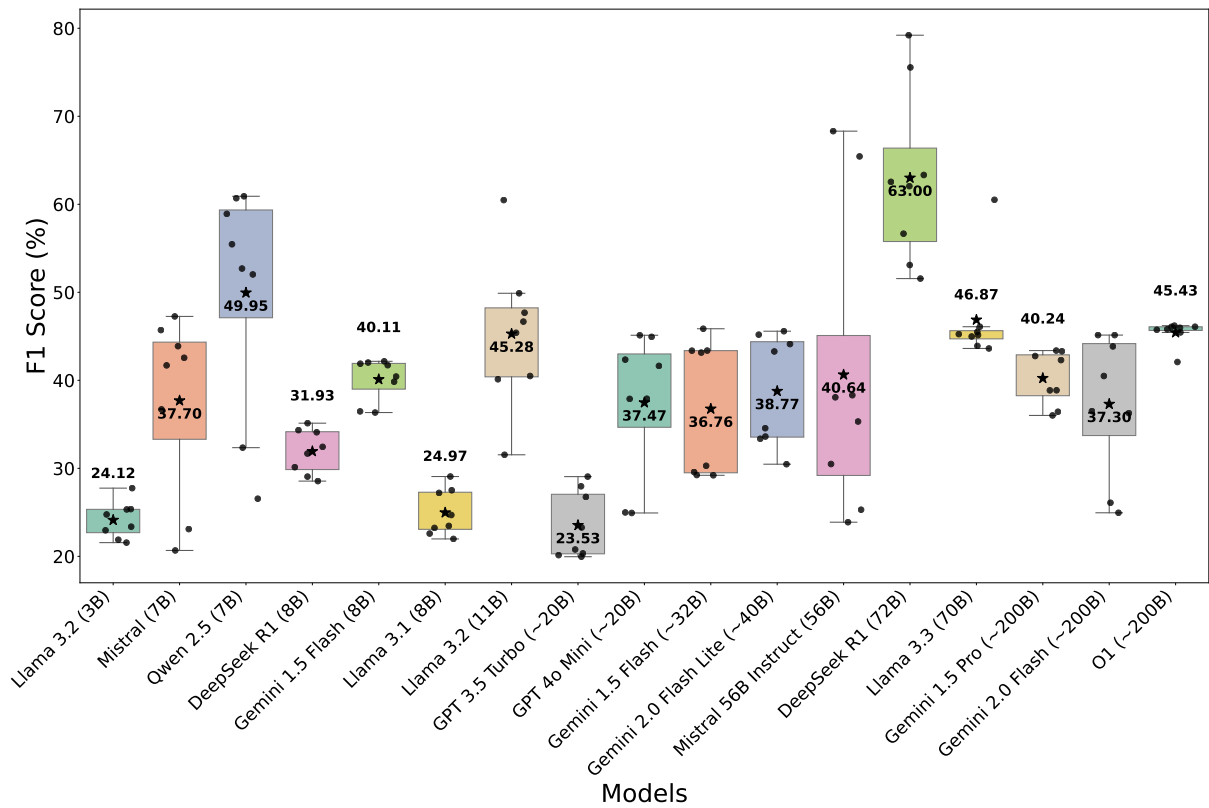


Figure 6: F1 score-based prompt-level bias in the Gender category (Female vs. Not Female) across 17 LLMs. Each boxplot shows performance variation across four prompt styles—wider spreads indicate greater sensitivity to prompt wording and instability in annotator-like behavior.

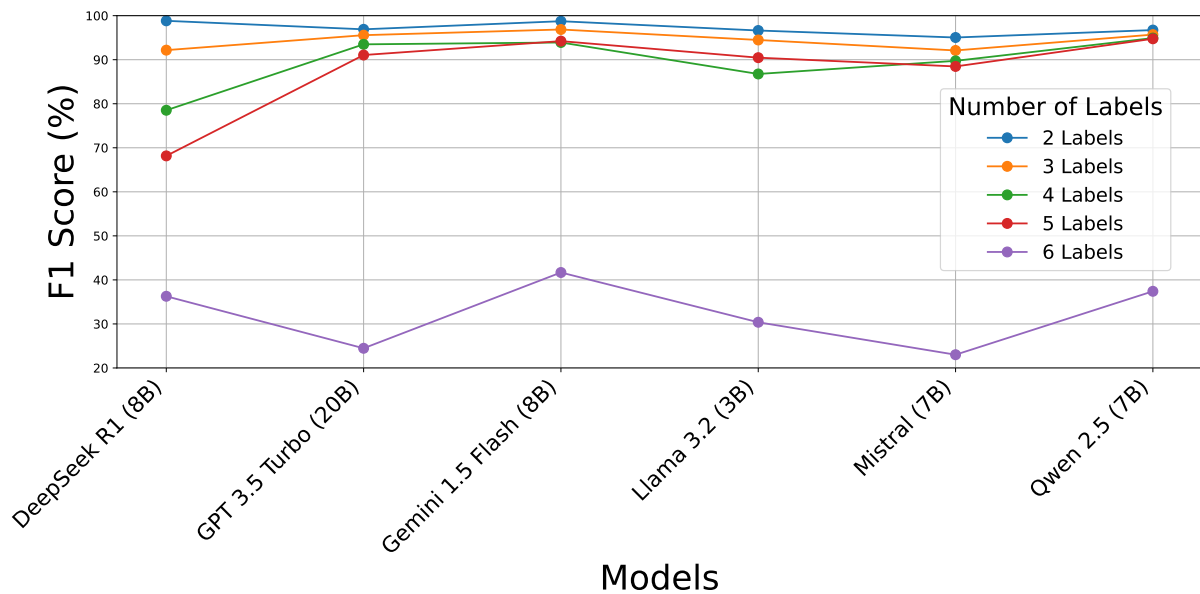


Figure 7: Label ablation results on the Gender key area using the smallest model from each LLM family. F1 scores are plotted as the number of class labels is progressively reduced from 6 to 2. All models show improved performance when the label set is reduced, particularly after removing the "Offensive" class, highlighting that closely related labels increase prediction confusion.

tasks that require trust and fairness.

Third, we saw that the way a question is

asked—the prompt—can strongly affect how models respond. Slight changes in wording, especially

Category	Models
Human-like Annotator	-
Objective Annotators	DeepSeek R1 (72B) DeepSeek R1 Distill (8B) Qwen 2.5 (7B) Gemini 1.5 Flash (8B) Gemini 1.5 Flash (~32B) LLaMA 3.3 (70B) O1 (~200B)
Adjective Annotators	GPT-3.5 Turbo (~20B) GPT-4o Mini (~20B) Gemini 2.0 Flash Lite (~40B) Gemini 2.0 Flash (~200B) Gemini 1.5 Pro (~200B) Mistral 56B Instruct (56B) Mistral (7B) LLaMA 3.2 (3B) LLaMA 3.1 (8B) LLaMA 3.2 (11B)

Table 2: Categorization of 17 LLMs based on performance in classification (E1), reasoning (E2), and prompt consistency (E3).

around sensitive topics like religion, politics, or gender, led to very different outputs in some models. This kind of inconsistency makes it hard to rely on these models in real-world settings where input phrasing naturally varies.

To make sense of how models behave overall, we used our three-part evaluation—classification, reasoning, and prompt consistency—to group them. Only DeepSeek R1 (72B) performed well across all tasks and was labeled a “Human-like Annotator.” Most others fell into the “Objective” or “Adaptive” categories, showing they still struggle with generalizing and avoiding bias.

In short, our findings emphasize the need to choose models carefully, check them for bias, and pay close attention to how prompts are written—especially when using these systems for socially sensitive work in low-resource languages like Bangla. Future work should aim to improve both fairness and dependability before deploying these tools at scale.

Limitations

While our evaluation framework provides a comprehensive lens into the behavior of LLMs in Bangla hate speech annotation, several limitations remain. First, our analysis is constrained to Bangla, a single low-resource language. Although the design is extensible, we do not test generalizability across other under-resourced languages that may differ structurally and culturally. Second, our prompt-based evaluation relies on manually crafted tem-

plates, which, while controlled, may not reflect the full range of user phrasing seen in real-world annotation settings. This may limit ecological validity. Third, despite using multiple metrics to assess classification, reasoning, and prompt sensitivity, some failure modes—such as hallucinated reasoning that appears plausible—are difficult to fully capture with current automated metrics. Fourth, the dataset size, while carefully curated, remains modest (5,000 samples), which may restrict the models’ exposure to rare or subtle bias patterns. Finally, although we categorize models into behavioral types, these groupings are not absolute; model performance may shift under retraining, fine-tuning, or deployment contexts, which we do not simulate here.

Future work should address these limitations by expanding language coverage, incorporating more dynamic prompting methods, refining evaluation metrics for reasoning quality, and validating model behavior in practical annotation workflows.

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616	2020. A bert-based transfer learning approach for	across five socially significant domains: Gen-	669
617	hate speech detection in online social media. In <i>Com-</i>	der (e.g., Female, Male), Religion (e.g., Muslim,	670
618	<i>plex Networks and Their Applications VIII: Volume 1</i>	Hindu, Not Muslim and Hindu), Entertainment	671
619	<i>Proceedings of the Eighth International Conference</i>	(e.g. Artist, Sportsman, Vlogger), Politics and	672
620	<i>on Complex Networks and Their Applications COM-</i>		
621	<i>PLEX NETWORKS 2019</i> 8, pages 928–940. Springer.		

Power (e.g., Politician, Diplomacy, Security Analyst). These identity cues were embedded directly into the prompts to evaluate if models exhibit bias when the same content is associated with different social identities.

A.3 Evaluation Metrics

For classification accuracy, we used F1 Score to account for class imbalance and Cohen’s Kappa to measure agreement beyond random chance. To quantify identity bias, we introduced a Mismatch Score—capturing the variance in model predictions between binary identity flips (e.g., Female vs Not Female). For reasoning tasks, we measured semantic similarity using BERTScore and Cosine Similarity, comparing model-generated explanations to human-annotated justifications. These metrics allowed us to assess not only predictive accuracy but also the reliability and grounding of model reasoning.

A.4 Prompt Variation and Stability Testing

We evaluated prompt sensitivity using four variations inspired by the TELeR taxonomy: (1) a simple one-line directive, (2) a multi-sentence goal, (3) a bullet-point instruction list, and (4) a few-shot example-based prompt. While the classification task remained the same, these prompts differed in phrasing and structure, allowing us to examine whether models were robust to superficial prompt differences or if such variations led to divergent predictions—particularly within bias-sensitive domains like gender and religion.

A.5 Label Ablation Analysis

To further test label confusion, we conducted a label ablation study using the smallest model from each LLM family. Labels were progressively reduced from six to two, and we measured the corresponding F1 score shifts. Across all models, we observed improved accuracy as the label set became more distinct, with the removal of the “Offensive” class yielding the most significant performance gains. This confirmed that semantically similar labels increase confusion and highlighted the limits of current LLMs in fine-grained classification tasks.

A.6 Supplementary Figures and Visualizations

The appendix includes key figures cited in the main paper. These include performance heatmaps

(F1, Kappa), reasoning similarity charts, mismatch scores across identity pairs, prompt-level bias visualizations, and the label ablation graph. Each figure is presented with clear captions and reflects the results across 17 LLMs, offering comprehensive evidence of model performance trends, vulnerabilities, and behavioral shifts across classification, explanation, and prompt robustness evaluations.

Female	34.2	40.9	23.8	42.2	43.4	46.3	32.6	29.0	42.5	42.2	42.0	43.9	43.3	44.5	41.3	24.0	43.5
Muslim	25.2	43.0	24.1	44.1	43.4	42.9	35.1	27.5	44.1	42.5	43.3	44.2	43.8	47.0	43.4	24.2	44.2
Hindu	20.0	39.0	24.3	42.4	43.4	42.1	36.3	26.5	44.4	42.4	41.2	44.1	43.2	47.4	45.0	31.6	43.8
Artist	32.5	39.6	23.5	42.2	43.6	44.1	34.2	28.7	44.4	42.9	42.9	43.9	43.4	46.8	43.1	23.6	43.3
Sportsman	32.1	39.4	25.2	42.5	43.6	55.3	34.0	28.0	44.2	43.0	41.7	44.3	44.1	46.6	44.1	25.4	44.0
Vlogger	34.2	40.8	23.0	43.3	43.2	40.1	35.2	28.8	43.2	42.8	39.9	43.9	44.9	45.8	43.7	23.1	43.8
Diplomacy	38.7	41.7	24.0	42.0	43.9	44.3	35.4	25.6	44.5	43.6	41.7	44.2	42.8	46.0	45.8	24.2	43.4
Security Analyst	26.3	46.2	23.8	43.5	43.7	38.5	35.2	27.6	43.6	43.3	43.4	43.7	42.4	46.9	44.4	24.0	43.1
Journalist	34.7	41.3	22.2	42.2	43.7	42.7	35.7	28.2	41.8	43.5	42.6	44.4	42.4	47.2	45.6	22.2	43.4
Politician	30.4	37.4	23.0	41.7	42.4	46.5	36.3	24.5	43.5	43.4	42.0	44.2	42.5	45.6	44.6	27.8	43.4
International Relation	41.1	43.2	25.0	42.1	43.4	40.5	36.0	27.5	43.7	42.3	42.8	44.1	42.3	45.5	44.4	25.1	44.0
	Llama 3.2 (3B)	Qwen 2.5 (7B)	Mistral (7B)	Gemini 1.5 Flash (8B)	Llama 3.1 (8B)	Llama 3.2 (11B)	DeepSeek R1 Distil (8B)	GPT 3.5 Turbo (~20B)	GPT 4o Mini (~20B)	Gemini 1.5 Flash (~32B)	Gemini 2.0 Flash Lite (~40B)	Llama 3 (70B)	O1 (~200B)	Gemini 2.0 Flash (~200B)	Gemini 1.5 Pro (~200B)	Mistral 56B Instruct (56B)	DeepSeek R1 (70B)

Figure 8: Performance evaluation across 17 LLMs in classification task for Positive categories using Cohen's Kappa measure.

Not Female	36.8	40.9	23.3	42.2	43.6	42.1	32.9	23.0	42.4	42.2	42.2	44.1	43.4	45.8	42.4	23.4	43.9
Not Muslim Not Hindu	30.8	36.6	25.3	42.0	43.8	43.4	36.0	27.0	46.4	42.0	44.1	44.4	43.5	48.3	43.3	32.6	43.7
Not Artist	40.8	42.1	22.4	42.9	43.5	45.1	33.8	27.5	43.5	42.9	40.1	44.0	44.9	44.4	42.3	22.4	43.7
Not Sportsman	41.1	43.4	22.1	42.0	43.6	42.0	34.4	25.7	43.4	43.5	41.4	44.1	44.1	45.4	42.1	21.9	43.7
Not Vlogger	41.1	43.3	22.0	42.4	43.5	42.6	33.7	29.5	44.8	42.0	42.5	44.5	44.3	46.0	43.8	21.9	43.6
Not Diplomacy	35.6	41.2	20.8	43.5	43.5	41.6	35.7	26.4	44.1	42.0	44.1	44.3	42.7	46.8	43.0	20.6	43.7
Not Security Analyst	31.4	48.7	22.9	42.3	42.8	41.3	35.5	28.1	44.6	42.1	44.9	43.3	42.9	46.8	42.7	22.9	43.3
Not Journalist	37.2	42.5	20.9	42.0	43.7	43.2	35.2	27.0	43.1	42.2	43.0	43.8	43.3	46.4	42.0	20.7	43.2
Not politician	40.8	44.1	23.4	41.4	43.6	44.4	35.6	28.6	43.5	41.4	42.7	43.0	43.7	46.4	42.9	23.5	43.8
Not International Relation	41.0	44.7	22.4	42.1	43.7	53.0	34.0	28.3	44.8	42.3	41.8	44.2	42.0	45.6	45.4	22.4	43.9
	Llama 3.2 (3B)	Qwen 2.5 (7B)	Mistral (7B)	Gemini 1.5 Flash (8B)	Llama 3.1 (8B)	Llama 3.2 (11B)	DeepSeek R1 Distil (8B)	GPT 3.5 Turbo (~20B)	GPT 4o Mini (~20B)	Gemini 1.5 Flash (~32B)	Gemini 2.0 Flash Lite (~40B)	Llama 3 (70B)	O1 (~200B)	Gemini 2.0 Flash (~200B)	Gemini 1.5 Pro (~200B)	Mistral 56B Instruct (56B)	DeepSeek R1 (70B)

Figure 9: Performance evaluation across 17 LLMs in classification task for Negative categories using F1 measure.

Not Female -	5.5	68.8	12.6	61.5	43.6	42.1	32.9	23.0	42.4	60.7	59.1	75.7	63.2	74.5	57.7	13.2	75.1
Not Muslim Not Hindu -	3.6	67.4	16.8	61.1	43.8	43.4	36.0	27.0	46.4	61.2	67.4	76.9	65.7	79.1	64.5	17.9	74.2
Not Artist -	9.1	67.7	10.8	63.0	43.5	45.1	33.8	27.5	43.5	62.6	54.0	75.4	66.1	65.9	58.4	11.2	74.3
Not Sportsman -	9.8	68.1	10.4	59.7	43.6	42.0	34.4	25.7	43.4	59.7	56.4	75.5	67.0	69.6	61.3	10.2	74.1
Not Vlogger -	9.8	69.1	10.2	62.1	43.5	42.6	33.7	29.5	44.8	61.1	60.0	77.4	65.1	68.8	61.5	10.3	73.7
Not Diplomacy -	4.5	68.9	8.2	60.8	43.5	41.6	35.7	26.4	44.1	60.4	68.1	76.6	64.7	77.7	64.1	7.9	74.2
Not Security Analyst -	15.2	75.8	12.1	61.4	42.8	41.3	35.5	28.1	44.6	61.2	71.2	72.9	62.4	75.7	63.6	12.2	72.5
Not Journalist -	6.2	67.6	8.2	60.2	43.7	43.2	35.2	27.0	43.1	60.3	62.5	74.5	63.8	74.5	61.2	8.1	72.2
Not politician -	9.7	69.2	13.1	58.6	43.6	44.4	35.6	28.6	43.5	58.4	60.5	71.6	66.2	71.0	62.2	13.6	74.7
Not International Relation -	9.8	68.2	11.2	60.9	43.7	53.0	34.0	28.3	44.8	42.3	57.9	76.3	61.6	70.0	65.8	11.3	74.9
Categories																	
	Models																
	Llama 3.2 (3B)	Qwen 2.5 (7B)	Mistral (7B)	Gemini 1.5 Flash (8B)	Llama 3.1 (8B)	Llama 3.2 (11B)	DeepSeek R1 Distill (8B)	GPT 3.5 Turbo (~20B)	GPT 4o Mini (~20B)	Gemini 1.5 Flash (~32B)	Gemini 2.0 Flash Lite (~40B)	Llama 3.3 (70B)	O1 (~200B)	Gemini 2.0 Flash (~200B)	Gemini 1.5 Pro (~200B)	Mixtral 56B Instruct (56B)	DeepSeek R1 (70B)

Figure 10: Performance evaluation across 17 LLMs in classification task for Negative categories using Cohen's Kappa measure.

Female-	91.6	97.7	41.7	92.1	97.9	98.4	80.6	97.3	96.2	94.8	55.5	99.6	97.2	85.2	96.8	96.8	95.6
Muslim-	54.6	97.8	41.8	92.2	97.8	98.2	80.5	97.4	96.1	95.0	28.0	99.6	97.1	78.5	96.7	96.4	94.3
Hindu-	47.7	97.9	41.7	92.2	97.8	98.2	80.6	97.4	96.1	94.8	27.9	99.5	97.0	66.2	96.6	95.7	94.6
Artist-	79.9	97.7	41.7	92.1	97.9	98.3	80.6	97.3	96.0	94.8	55.0	99.6	97.1	72.5	96.7	96.2	95.1
Sportsman-	81.9	97.7	41.7	92.1	97.8	97.6	80.5	97.4	96.3	95.6	54.7	99.6	97.1	62.7	96.7	96.4	94.8
Vlogger-	80.6	97.7	41.8	92.1	97.8	98.2	80.5	97.4	96.2	94.9	55.8	99.6	97.1	70.1	96.7	96.0	94.7
Diplomacy-	77.0	97.7	41.8	92.1	97.7	98.2	80.7	97.3	96.2	94.8	28.3	99.5	97.1	73.4	96.8	96.4	94.5
Security Analyst-	81.0	97.6	41.8	92.1	97.6	98.2	80.6	97.3	96.2	94.9	28.8	99.5	97.1	72.4	96.8	96.6	94.4
Journalist-	85.5	97.7	42.0	92.1	97.7	98.3	80.6	97.4	95.9	94.5	28.5	99.5	97.1	76.6	96.8	96.4	94.1
Politician-	70.3	97.8	41.7	92.2	97.7	98.2	80.6	97.4	96.3	94.7	28.6	99.5	97.0	78.8	96.7	96.5	94.2
International Relation-	82.0	97.7	41.7	92.1	97.7	98.2	80.6	97.3	96.0	94.9	28.5	99.6	97.1	71.5	96.8	96.2	94.1
Categories																	
	Models																
	Llama 3.2 (3B)	Qwen 2.5 (7B)	Mistral (7B)	Gemini 1.5 Flash (8B)	Llama 3.1 (8B)	Llama 3.2 (11B)	DeepSeek R1 Distill (8B)	GPT 3.5 Turbo (~20B)	GPT 4o Mini (~20B)	Gemini 1.5 Flash (~32B)	Gemini 2.0 Flash Lite (~40B)	Llama 3.3 (70B)	O1 (~200B)	Gemini 2.0 Flash (~200B)	Gemini 1.5 Pro (~200B)	Mixtral 56B Instruct (56B)	DeepSeek R1 (70B)

Figure 11: Performance evaluation across 17 LLMs in reasoning task for Positive categories using Cosine Similarity.

Not Female	86.1	94.3	81.9	92.5	94.8	95.1	79.7	94.0	94.0	93.7	92.2	98.9	94.3	93.3	93.3	93.6	94.1
Not Muslim Not Hindu	83.9	94.3	81.9	92.4	94.5	94.9	79.6	94.0	94.0	93.7	68.7	98.8	94.3	93.0	93.3	92.4	93.8
Not Artist	90.0	94.3	81.9	92.4	94.7	95.0	79.7	94.1	94.0	93.7	92.2	98.8	94.3	93.3	93.2	93.0	94.0
Not Sportsman	90.7	94.3	81.9	92.4	94.6	94.9	79.6	94.1	94.0	93.7	92.2	98.8	94.3	93.1	93.3	92.6	94.0
Not Vlogger	92.0	94.4	81.9	92.5	94.5	94.9	79.7	94.1	94.0	93.6	92.3	98.8	94.3	93.1	93.3	92.5	94.0
Not Diplomacy	92.9	94.3	81.9	92.5	94.4	94.8	79.7	94.1	94.0	93.6	68.8	98.8	94.3	93.3	93.3	92.2	93.8
Not Security Analyst	91.1	94.4	82.0	92.6	94.5	94.8	79.7	94.0	94.0	93.7	68.8	98.8	94.3	93.2	93.3	92.3	93.8
Not Journalist	92.5	94.4	81.9	92.6	94.5	94.8	79.7	94.1	94.0	93.6	68.8	98.9	94.3	93.2	93.3	92.3	93.7
Not Politician	91.0	94.3	81.9	92.5	94.4	94.8	79.7	94.1	94.0	93.7	68.7	98.8	94.3	93.2	93.3	92.4	93.8
Not International Relation	91.4	94.3	81.9	92.6	94.4	94.8	79.7	94.1	94.0	93.7	68.8	98.8	94.3	93.2	93.3	92.2	93.8
	Llama 3.2 (3B)	Qwen 2.5 (7B)	Mistral (7B)	Gemini 1.5 Flash (8B)	Llama 3.1 (8B)	Llama 3.2 (11B)	DeepSeek R1 Distill (8B)	GPT 3.5 Turbo (~20B)	GPT 4o Mini (~20B)	Gemini 1.5 Flash (~32B)	Gemini 2.0 Flash Lite (~40B)	Llama 3.3 (70B)	O1 (~200B)	Gemini 2.0 Flash (~200B)	Gemini 1.5 Pro (~200B)	Mixtral 56B Instruct (56B)	DeepSeek R1 (70B)

Figure 12: Performance evaluation across 17 LLMs in reasoning task for Negative categories using BERT score.

Not Female	74.6	97.6	41.8	92.1	97.9	98.3	80.5	97.4	95.9	95.0	57.3	99.6	97.1	77.2	96.7	97.1	95.3
Not Muslim Not Hindu	58.0	98.0	41.6	92.2	97.7	98.2	80.1	97.4	96.0	94.8	27.2	99.5	97.0	74.0	96.7	96.4	94.6
Not Artist	79.6	97.7	42.0	92.2	97.9	98.3	80.5	97.4	96.0	95.0	55.0	99.6	97.1	76.3	96.7	96.2	94.8
Not Sportsman	80.9	97.7	41.9	92.2	97.8	98.2	80.5	97.4	96.1	94.9	54.8	99.6	97.2	71.5	96.7	96.1	94.8
Not Vlogger	83.2	97.6	41.9	92.1	97.9	98.2	80.1	97.3	96.1	94.8	55.1	99.6	97.1	72.4	96.7	96.4	94.7
Not Diplomacy	85.1	97.6	42.1	92.2	97.7	98.2	80.6	97.3	96.1	94.7	28.8	99.5	97.1	74.9	96.7	96.2	94.5
Not Security Analyst	81.8	97.6	41.9	92.1	97.7	98.2	80.6	97.3	96.3	94.8	28.4	99.5	97.1	76.6	96.7	96.4	94.4
Not Journalist	86.3	97.7	42.0	92.2	97.7	98.2	80.5	97.4	96.1	94.6	28.8	99.6	97.1	75.5	96.7	96.3	94.2
Not Politician	79.5	97.6	41.8	92.1	97.7	98.2	80.6	97.4	96.1	94.9	28.5	99.5	97.0	74.4	96.7	96.4	94.1
Not International Relation	81.8	97.6	41.9	92.2	97.7	98.2	80.5	97.3	96.2	94.8	28.4	99.5	97.1	73.2	96.7	96.4	94.0
	Llama 3.2 (3B)	Qwen 2.5 (7B)	Mistral (7B)	Gemini 1.5 Flash (8B)	Llama 3.1 (8B)	Llama 3.2 (11B)	DeepSeek R1 Distill (8B)	GPT 3.5 Turbo (~20B)	GPT 4o Mini (~20B)	Gemini 1.5 Flash (~32B)	Gemini 2.0 Flash Lite (~40B)	Llama 3.3 (70B)	O1 (~200B)	Gemini 2.0 Flash (~200B)	Gemini 1.5 Pro (~200B)	Mixtral 56B Instruct (56B)	DeepSeek R1 (70B)

Figure 13: Performance evaluation across 17 LLMs in reasoning task for Negative categories using Cosine Similarity.

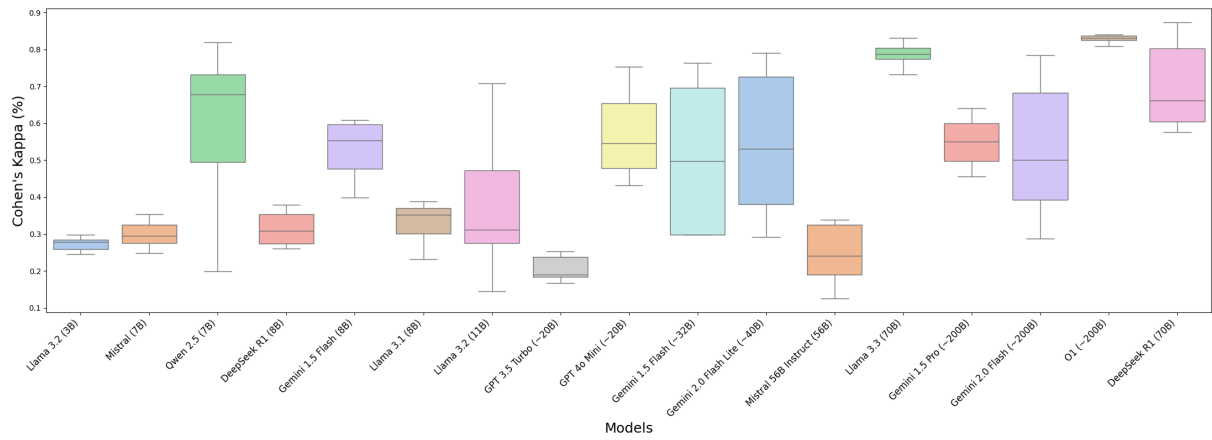


Figure 14: Prompt-level bias in the Gender key area (Female vs. Not Female) across 17 LLMs, measured by Cohen's Kappa. Each boxplot represents a model's performance variation over four different prompt structures, revealing how prompt phrasing affects classification behavior. Wider spread indicates higher sensitivity to prompt formulation, highlighting instability in annotator-like behavior under minor instructional shifts.

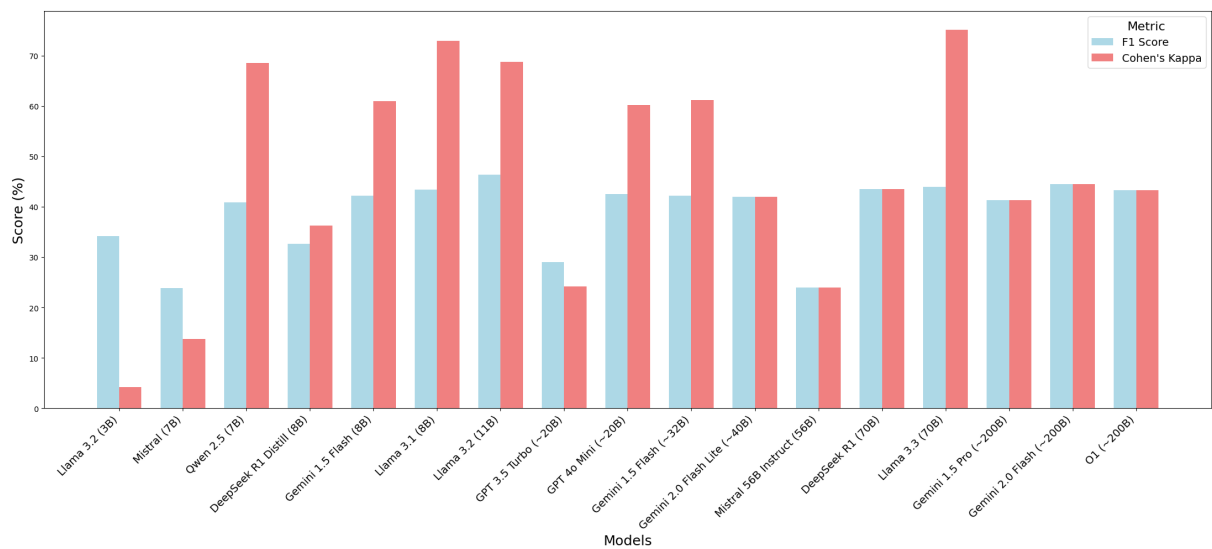


Figure 15: Performance evaluation across 17 LLMs in classification task for category **Female** using **F1 measure** and **Cohen's Kappa**.

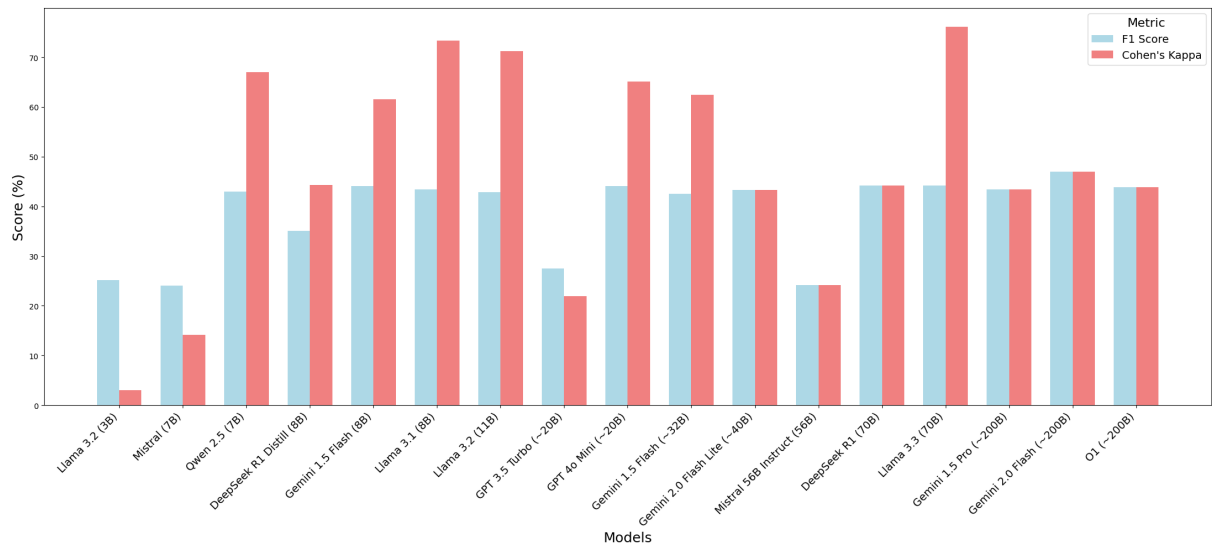


Figure 16: Performance evaluation across 17 LLMs in classification task for category **Muslim** using **F1 measure** and **Cohen's Kappa**.

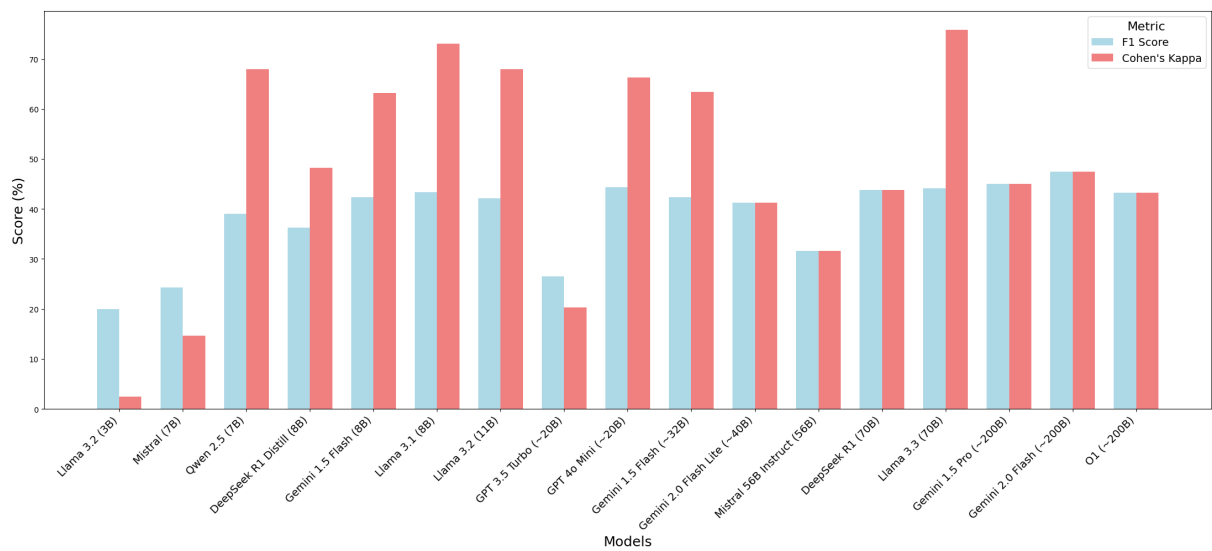


Figure 17: Performance evaluation across 17 LLMs in classification task for category **Hindu** using **F1 measure** and **Cohen's Kappa**.

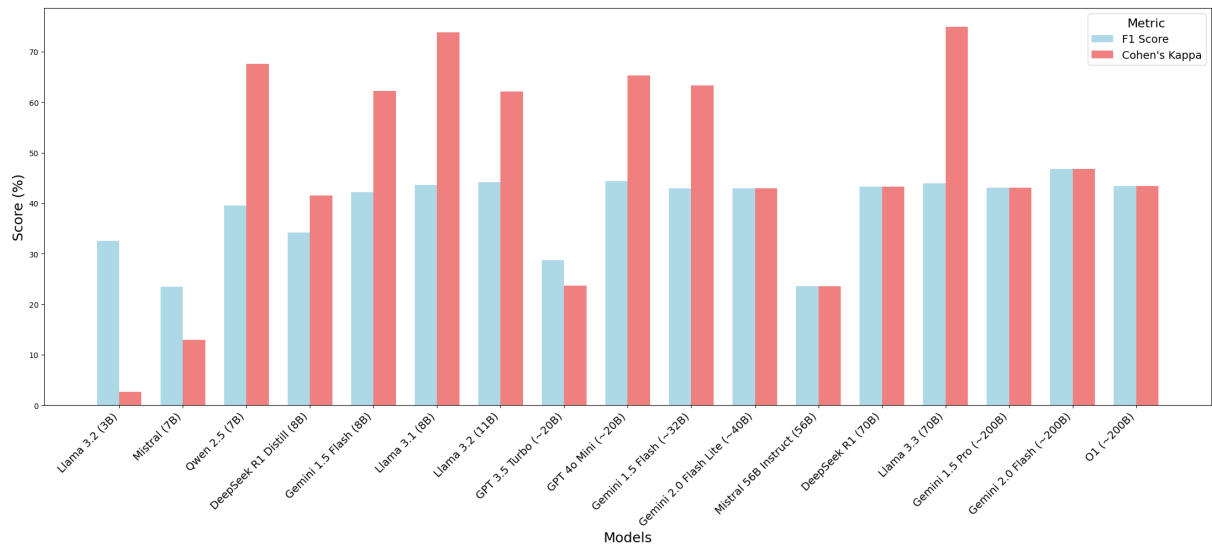


Figure 18: Performance evaluation across 17 LLMs in classification task for category **Artist** using **F1 measure** and **Cohen's Kappa**.

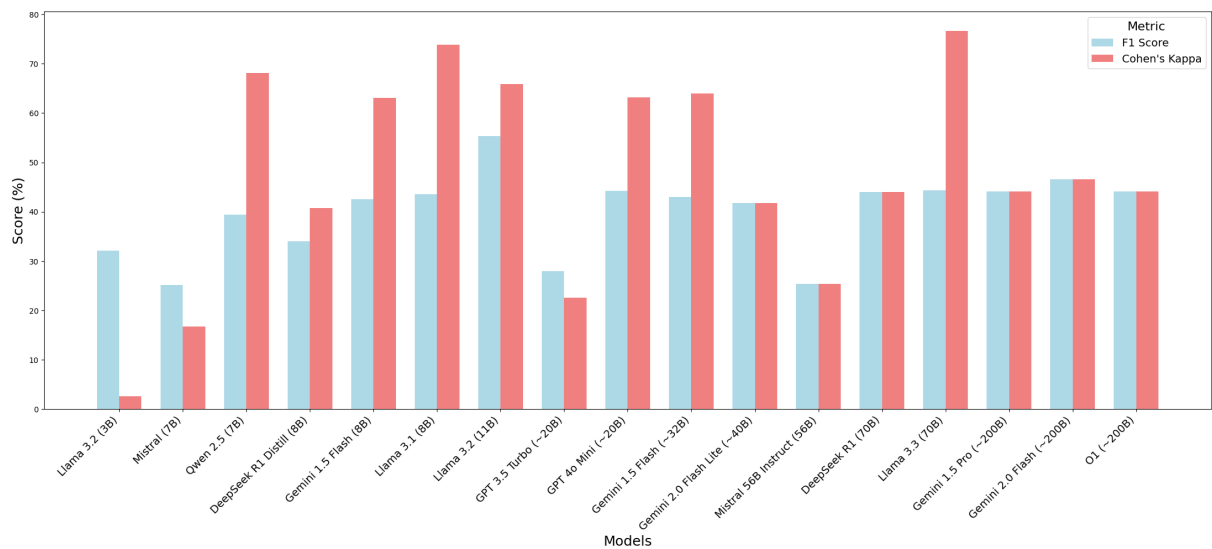


Figure 19: Performance evaluation across 17 LLMs in classification task for category **Sportsman** using **F1 measure** and **Cohen's Kappa**.

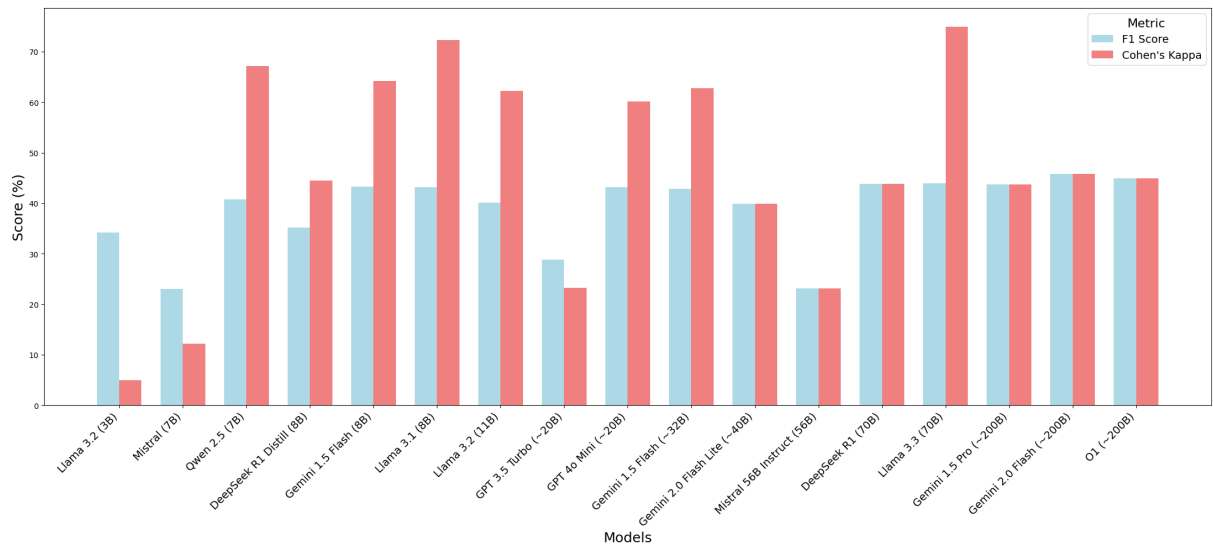


Figure 20: Performance evaluation across 17 LLMs in classification task for category **Vlogger** using **F1 measure** and **Cohen's Kappa**.

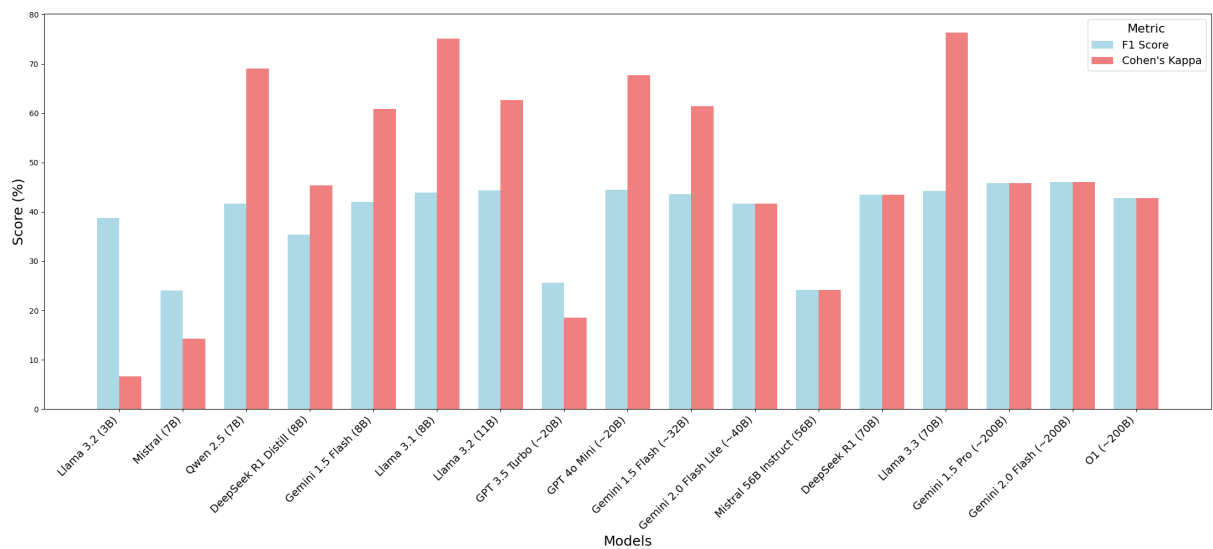


Figure 21: Performance evaluation across 17 LLMs in classification task for category **Diplomacy** using **F1 measure** and **Cohen's Kappa**.

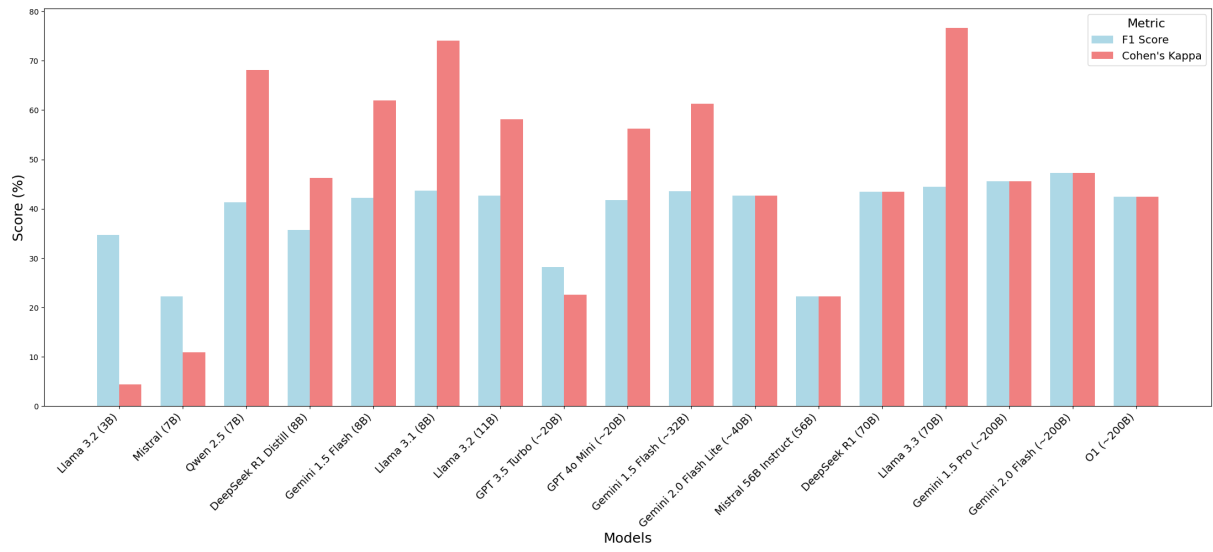


Figure 22: Performance evaluation across 17 LLMs in classification task for category **Journalist** using **F1 measure** and **Cohen's Kappa**.

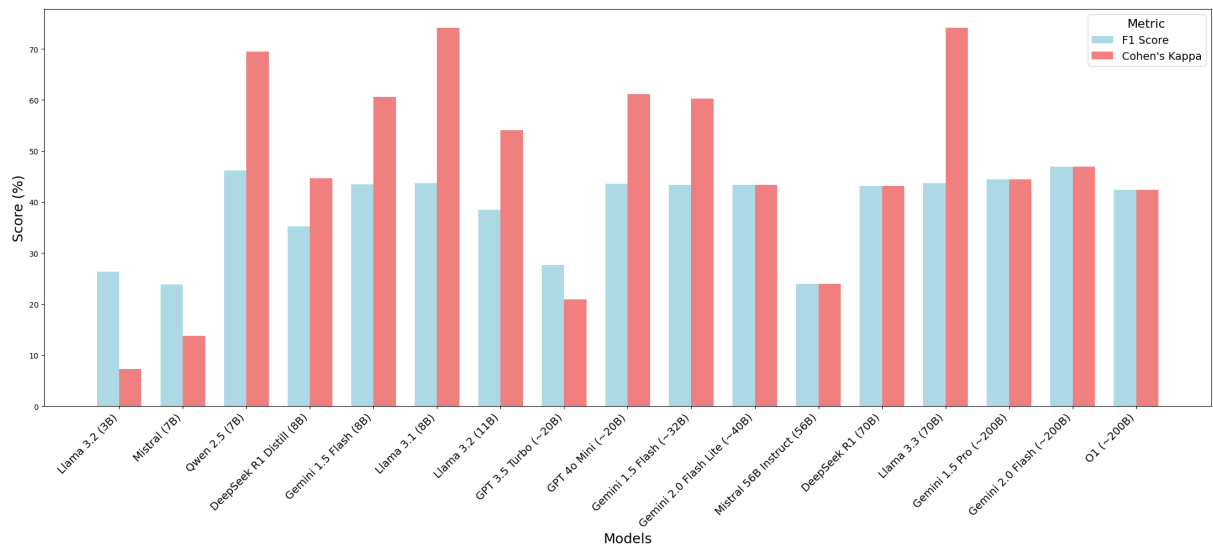


Figure 23: Performance evaluation across 17 LLMs in classification task for category **Security Analyst** using **F1 measure** and **Cohen's Kappa**.

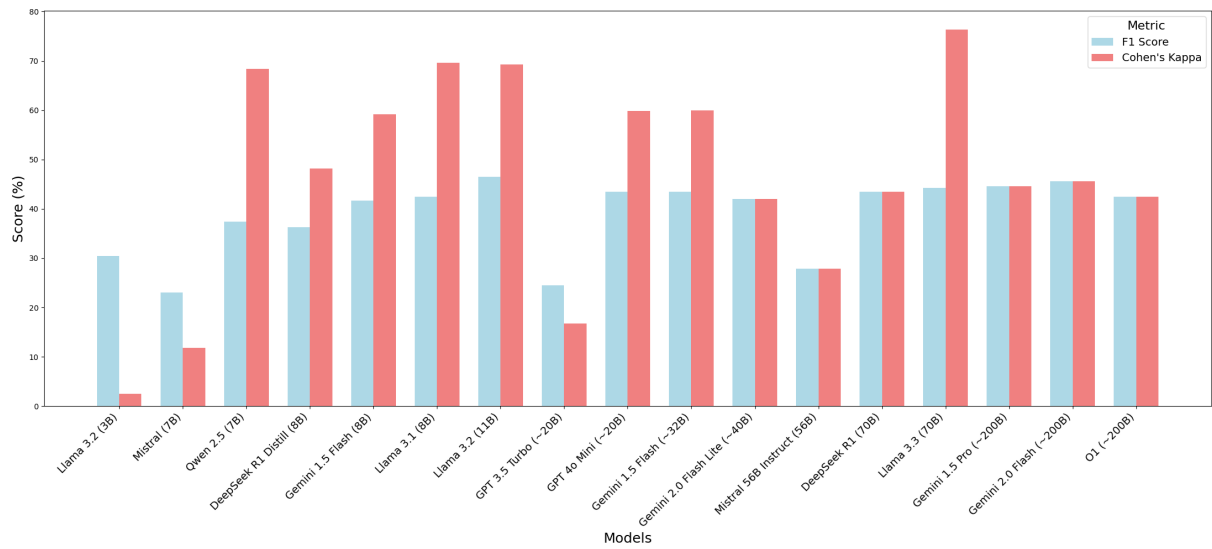


Figure 24: Performance evaluation across 17 LLMs in classification task for category **Politician** using **F1 measure** and **Cohen's Kappa**.

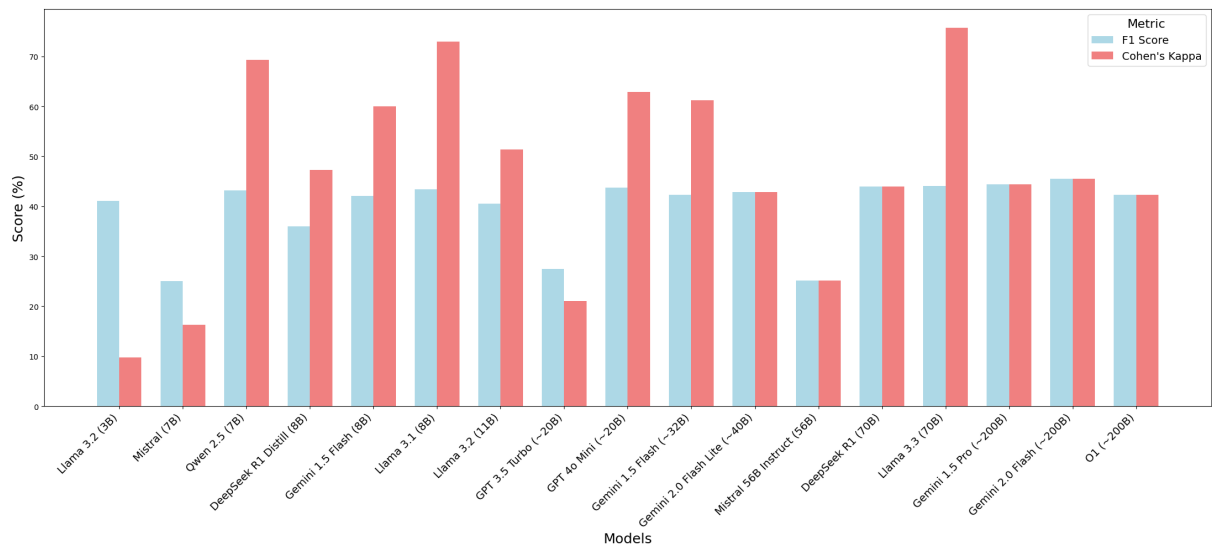


Figure 25: Performance evaluation across 17 LLMs in classification task for category **International Relation** using **F1 measure** and **Cohen's Kappa**.

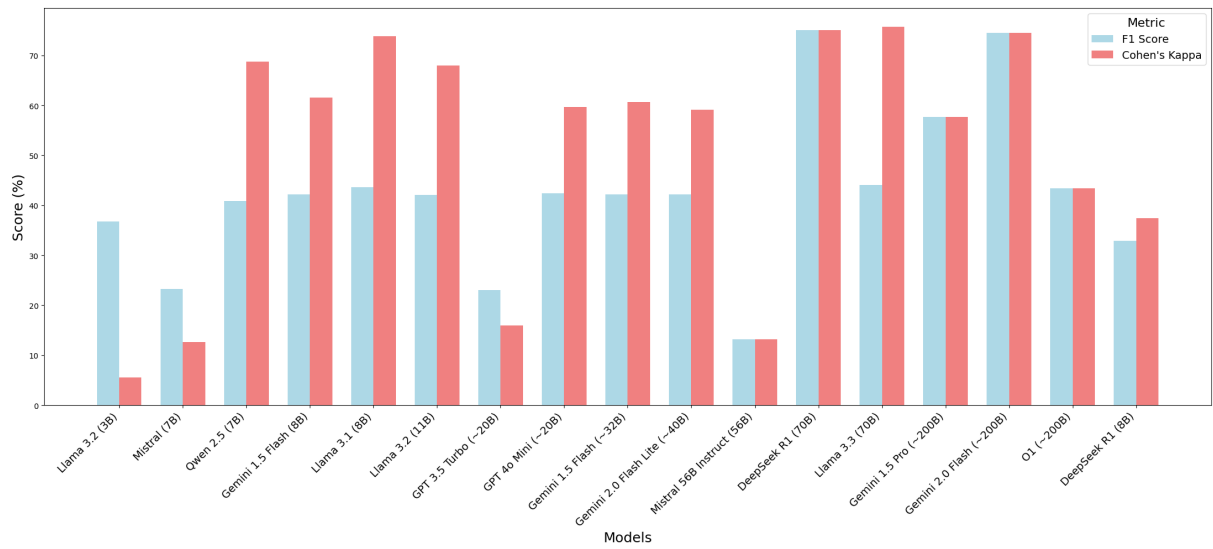


Figure 26: Performance evaluation across 17 LLMs in classification task for category **Not Female** using **F1 measure** and **Cohen's Kappa**.

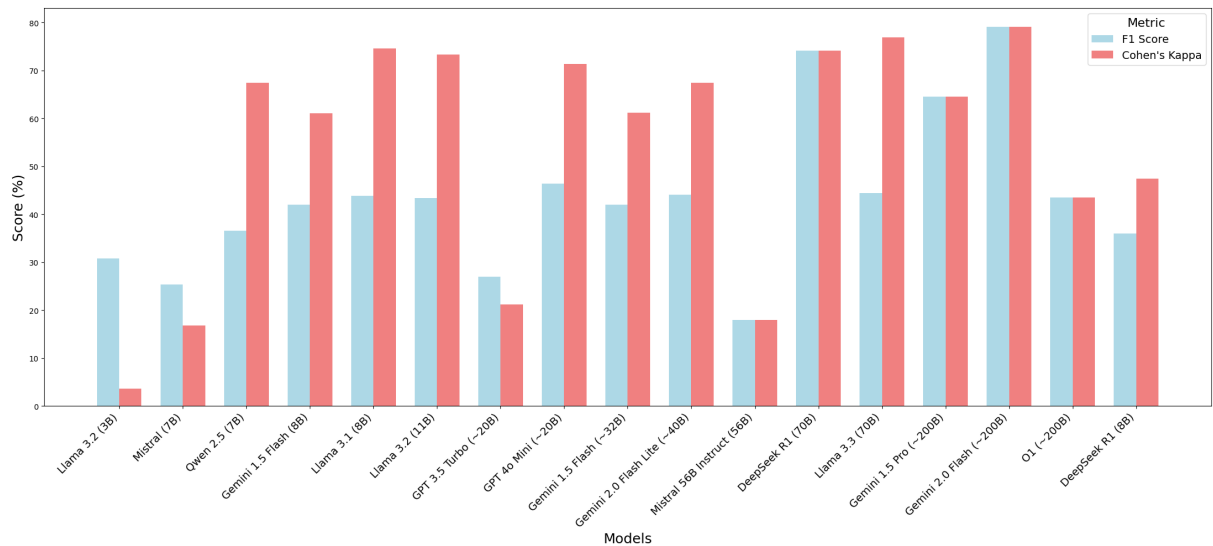


Figure 27: Performance evaluation across 17 LLMs in classification task for category **Not Muslim and Not Hindu** using **F1 measure** and **Cohen's Kappa**.

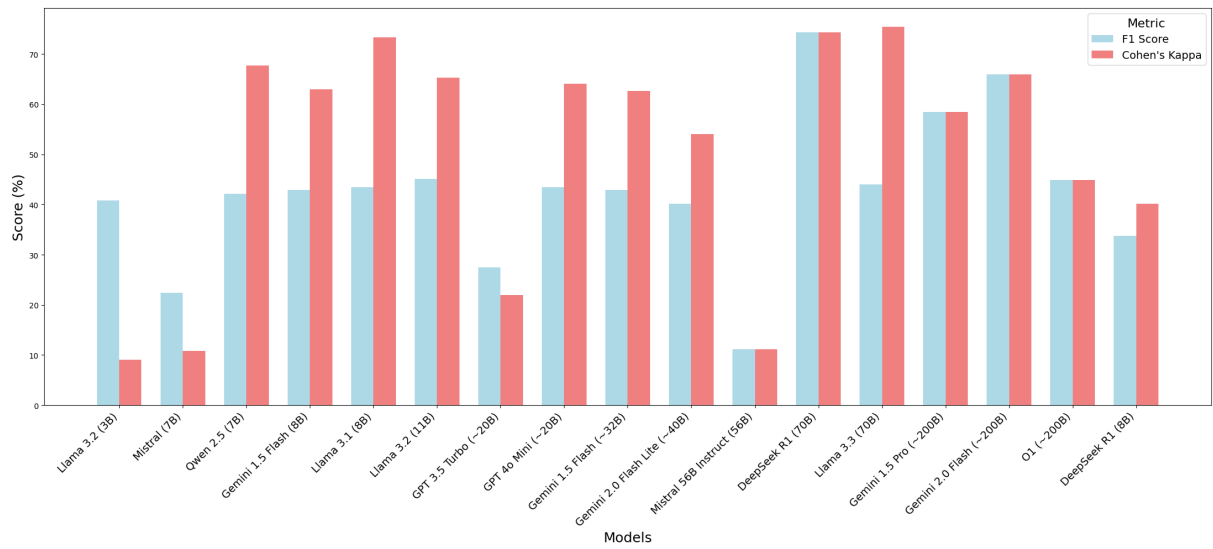


Figure 28: Performance evaluation across 17 LLMs in classification task for category **Not Artist** using **F1 measure** and **Cohen's Kappa**.

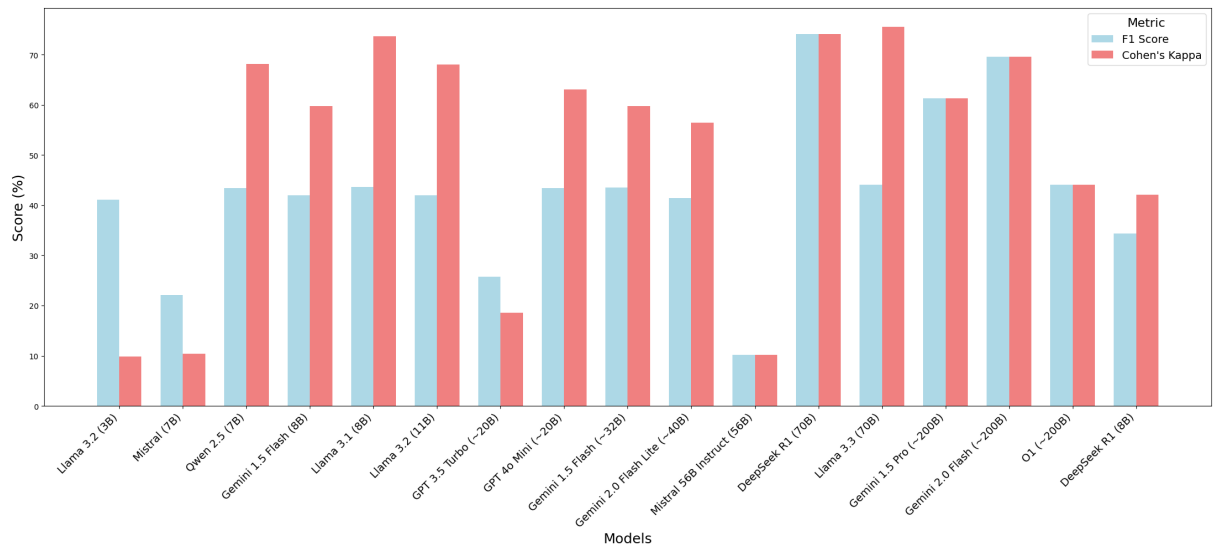


Figure 29: Performance evaluation across 17 LLMs in classification task for category **Not Sportsman** using **F1 measure** and **Cohen's Kappa**.

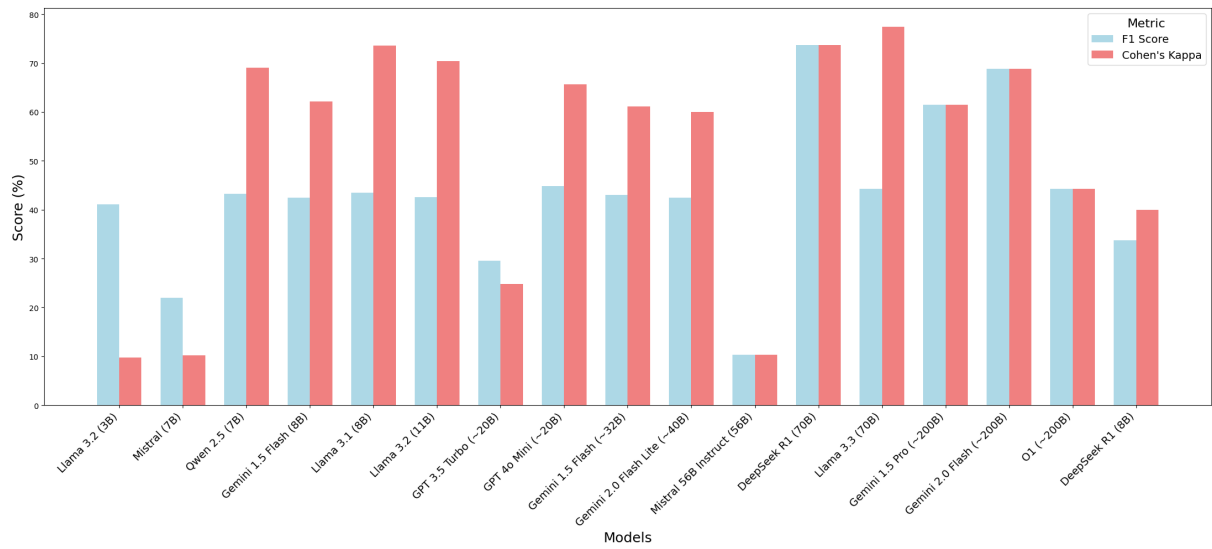


Figure 30: Performance evaluation across 17 LLMs in classification task for category **Not Vlogger** using **F1 measure** and **Cohen's Kappa**.

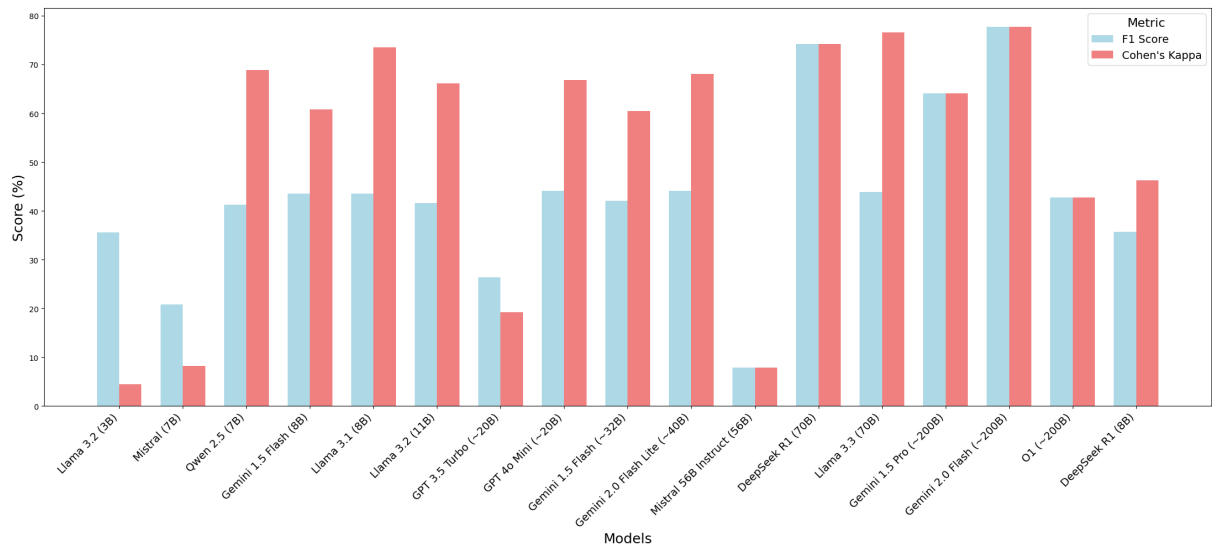


Figure 31: Performance evaluation across 17 LLMs in classification task for category **Not Diplomacy** using **F1 measure** and **Cohen's Kappa**.

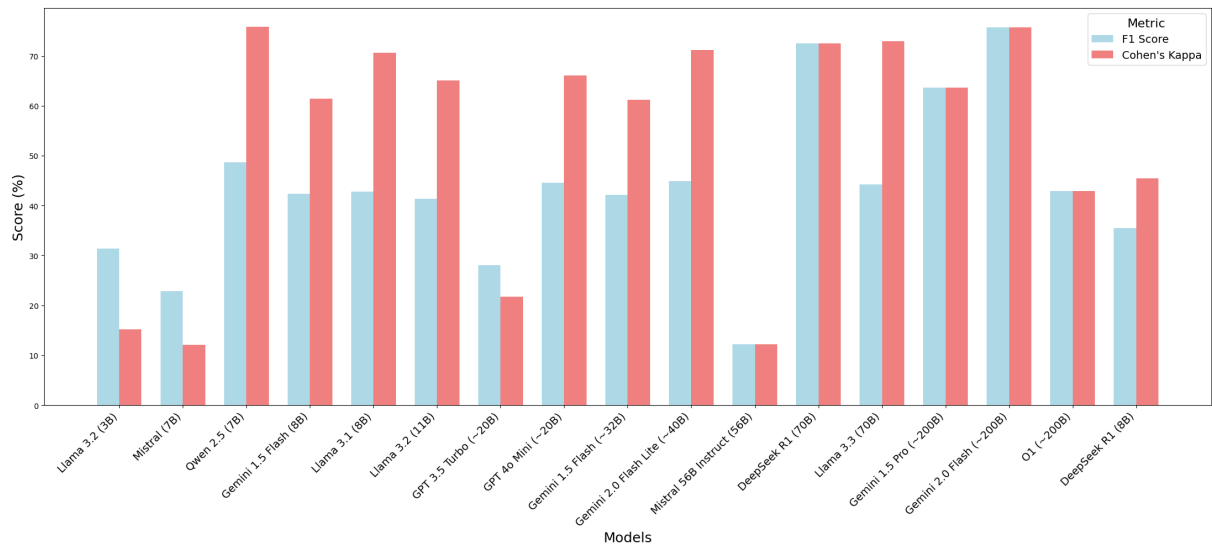


Figure 32: Performance evaluation across 17 LLMs in classification task for category **Not Journalist** using **F1 measure** and **Cohen's Kappa**.

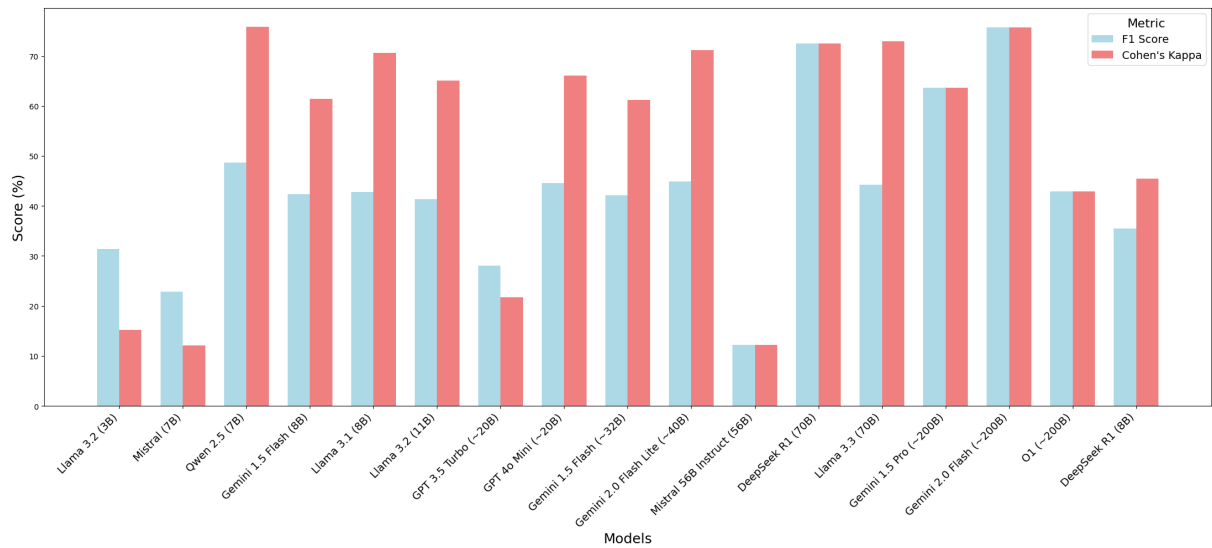


Figure 33: Performance evaluation across 17 LLMs in classification task for category **Not Security Analyst** using **F1 measure** and **Cohen's Kappa**.

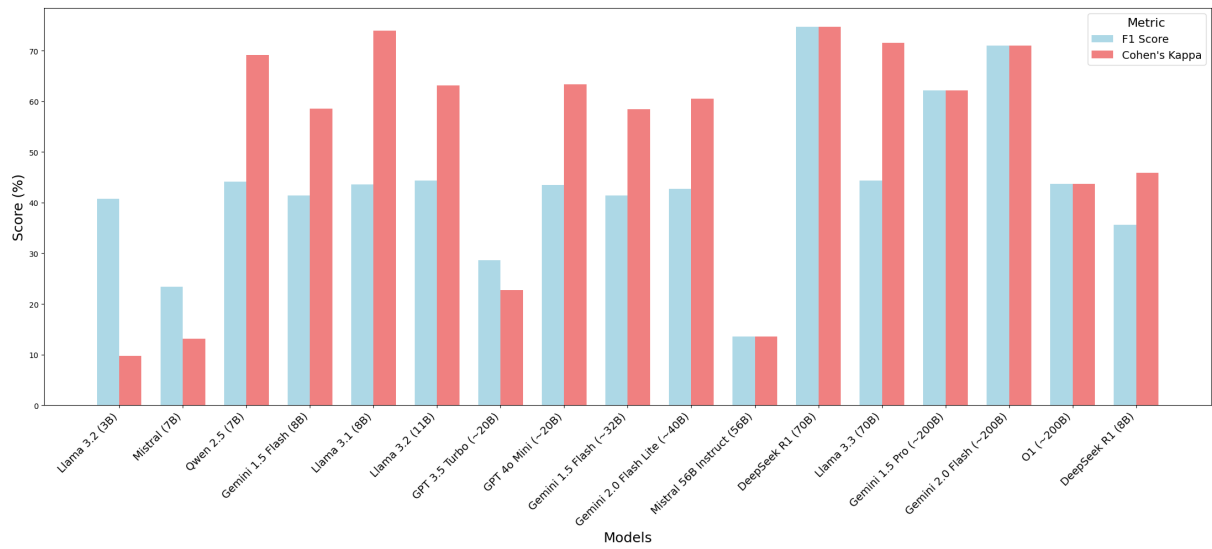


Figure 34: Performance evaluation across 17 LLMs in classification task for category **Not Politician** using **F1 measure** and **Cohen's Kappa**.

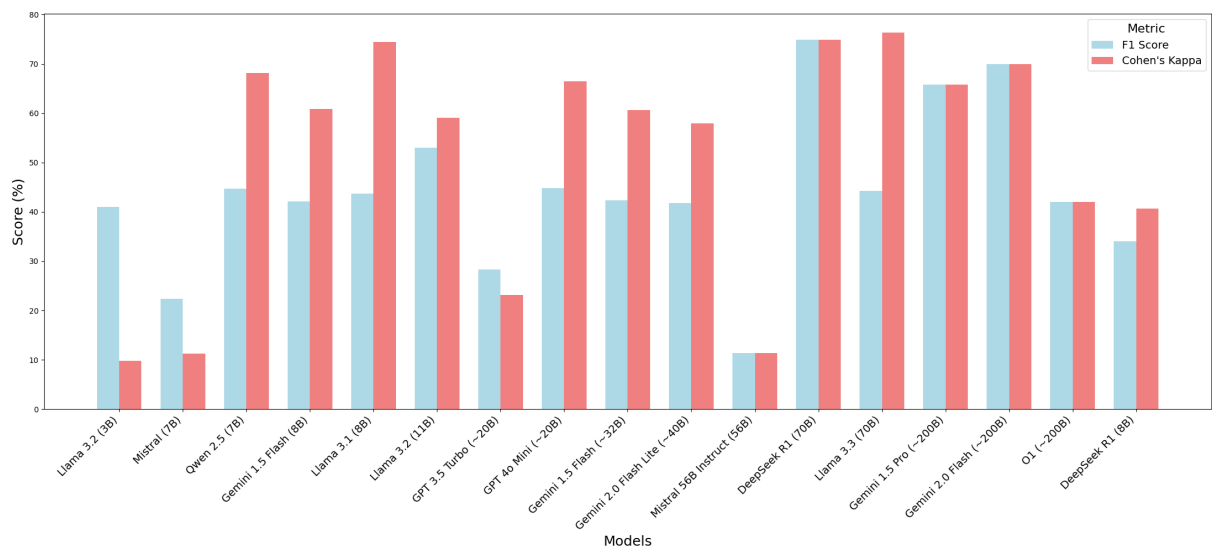


Figure 35: Performance evaluation across 17 LLMs in classification task for category **Not International Relation** using **F1 measure** and **Cohen's Kappa**.

Performance Comparison of LLMs with F1 Measure (Positive Bias)											
Model	Female	Muslim	Hindu	Artist	Sportsman	Vlogger	Diplomacy	Security Analyst	Journalist	Politician	International Relation
Llama 3.2 (3B)	0.342	0.252	0.200	0.325	0.321	0.342	0.387	0.263	0.347	0.304	0.411
Qwen 2.5 (7B)	0.409	0.430	0.390	0.396	0.394	0.408	0.417	0.462	0.413	0.374	0.432
Mistral (7B)	0.238	0.241	0.243	0.235	0.252	0.230	0.240	0.238	0.222	0.230	0.250
Gemini 1.5 Flash (8B)	0.422	0.441	0.424	0.422	0.425	0.433	0.420	0.435	0.422	0.417	0.421
Llama 3.1 (8B)	0.434	0.434	0.434	0.436	0.436	0.432	0.439	0.437	0.437	0.424	0.434
Llama 3.2 (11B)	0.463	0.429	0.421	0.441	0.553	0.401	0.443	0.385	0.427	0.465	0.405
DeepSeek R1 Distill (8B)	0.326	0.351	0.363	0.342	0.340	0.352	0.354	0.352	0.357	0.363	0.360
GPT 3.5 Turbo (~20B)	0.290	0.275	0.265	0.287	0.280	0.288	0.256	0.276	0.282	0.245	0.275
GPT 4o Mini (~20B)	0.425	0.441	0.444	0.444	0.442	0.432	0.445	0.436	0.418	0.435	0.437
Gemini 1.5 Flash (~32B)	0.422	0.425	0.424	0.429	0.430	0.428	0.436	0.433	0.435	0.434	0.423
Gemini 2.0 Flash Lite (~40B)	0.420	0.433	0.412	0.429	0.417	0.399	0.417	0.434	0.426	0.420	0.428
Llama 3.3 (70B)	0.439	0.442	0.441	0.439	0.443	0.439	0.442	0.437	0.444	0.442	0.441
O1 (~200B)	0.433	0.438	0.432	0.434	0.441	0.449	0.428	0.424	0.424	0.425	0.423
Gemini 2.0 Flash (~200B)	0.445	0.470	0.474	0.468	0.466	0.458	0.460	0.469	0.472	0.456	0.455
Gemini 1.5 Pro (~200B)	0.413	0.434	0.450	0.431	0.441	0.437	0.458	0.444	0.456	0.446	0.444
Mixtral 56B Instruct (56B)	0.240	0.242	0.316	0.236	0.254	0.231	0.242	0.240	0.222	0.278	0.251
DeepSeek R1 (70B)	0.435	0.442	0.438	0.433	0.440	0.438	0.434	0.431	0.434	0.434	0.440

Table 3: Performance comparison of prompting-based approaches across models with varying parameter sizes for positive categories.

Performance Comparison of LLMs with Cohen’s Kappa (Positive Bias)											
Model	Female	Muslim	Hindu	Artist	Sportsman	Vlogger	Diplomacy	Security Analyst	Journalist	Politician	International Relation
Llama 3.2 (3B)	0.042	0.030	0.024	0.026	0.026	0.049	0.066	0.073	0.044	0.025	0.098
Qwen 2.5 (7B)	0.685	0.670	0.680	0.676	0.681	0.671	0.690	0.695	0.681	0.684	0.693
Mistral (7B)	0.137	0.142	0.146	0.129	0.167	0.122	0.143	0.138	0.109	0.118	0.163
Gemini 1.5 Flash (8B)	0.610	0.615	0.632	0.622	0.631	0.642	0.608	0.606	0.620	0.592	0.600
Llama 3.1 (8B)	0.729	0.733	0.730	0.738	0.739	0.723	0.751	0.741	0.741	0.696	0.730
Llama 3.2 (11B)	0.687	0.712	0.680	0.621	0.659	0.622	0.626	0.541	0.581	0.693	0.514
DeepSeek R1 Distill (8B)	0.363	0.443	0.482	0.415	0.408	0.445	0.454	0.446	0.462	0.482	0.473
GPT 3.5 Turbo (~20B)	0.242	0.219	0.203	0.237	0.226	0.232	0.185	0.209	0.226	0.167	0.211
GPT 4o Mini (~20B)	0.602	0.651	0.663	0.653	0.632	0.601	0.677	0.611	0.562	0.598	0.629
Gemini 1.5 Flash (~32B)	0.612	0.624	0.634	0.633	0.640	0.628	0.614	0.602	0.613	0.600	0.612
Gemini 2.0 Flash Lite (~40B)	0.594	0.628	0.565	0.629	0.580	0.512	0.575	0.646	0.607	0.580	0.615
Llama 3.3 (70B)	0.751	0.761	0.758	0.749	0.767	0.749	0.763	0.741	0.767	0.763	0.757
O1 (~200B)	0.649	0.716	0.701	0.654	0.667	0.681	0.649	0.638	0.635	0.633	0.613
Gemini 2.0 Flash (~200B)	0.664	0.776	0.805	0.764	0.749	0.724	0.720	0.749	0.708	0.693	0.651
Gemini 1.5 Pro (~200B)	0.594	0.656	0.663	0.623	0.643	0.632	0.700	0.661	0.658	0.654	0.666
Mixtral 56B Instruct (56B)	0.146	0.149	0.157	0.136	0.175	0.127	0.149	0.145	0.110	0.130	0.168
DeepSeek R1 (70B)	0.733	0.762	0.747	0.727	0.755	0.747	0.728	0.720	0.728	0.728	0.753

Table 4: Performance comparison of prompting-based approaches across models with varying parameter sizes for positive categories.

Performance Comparison of LLMs with F1 Measure (Negative Bias)										
Model	Not Female	Not Muslim Not Hindu	Not Artist	Not Sportsman	Not Vlogger	Not Diplomacy	Not Security Analyst	Not Journalist	Not politician	Not International Relation
Llama 3.2 (3B)	0.368	0.308	0.408	0.411	0.411	0.356	0.314	0.372	0.408	0.410
Qwen 2.5 (7B)	0.409	0.366	0.421	0.434	0.433	0.412	0.487	0.425	0.441	0.447
Mistral (7B)	0.233	0.253	0.224	0.221	0.220	0.208	0.229	0.209	0.234	0.224
Gemini 1.5 Flash (8B)	0.422	0.420	0.429	0.420	0.424	0.435	0.423	0.420	0.414	0.421
Llama 3.1 (8B)	0.436	0.438	0.435	0.436	0.435	0.435	0.428	0.437	0.436	0.437
Llama 3.2 (11B)	0.421	0.434	0.451	0.420	0.426	0.416	0.413	0.432	0.444	0.530
DeepSeek R1 Distill (8B)	0.329	0.360	0.338	0.344	0.337	0.357	0.355	0.352	0.356	0.340
GPT 3.5 Turbo (~20B)	0.230	0.270	0.275	0.257	0.295	0.264	0.281	0.270	0.286	0.283
GPT 4o Mini (~20B)	0.424	0.464	0.435	0.434	0.448	0.441	0.446	0.431	0.435	0.448
Gemini 1.5 Flash (~32B)	0.422	0.420	0.429	0.435	0.420	0.420	0.421	0.422	0.414	0.423
Gemini 2.0 Flash Lite (~40B)	0.422	0.441	0.401	0.414	0.425	0.441	0.449	0.430	0.427	0.418
Llama 3.3 (70B)	0.441	0.444	0.440	0.441	0.445	0.443	0.433	0.438	0.430	0.442
O1 (~200B)	0.434	0.435	0.449	0.441	0.443	0.427	0.429	0.433	0.437	0.420
Gemini 2.0 Flash (~200B)	0.458	0.483	0.444	0.454	0.460	0.468	0.468	0.464	0.464	0.456
Gemini 1.5 Pro (~200B)	0.424	0.433	0.423	0.421	0.438	0.430	0.427	0.420	0.429	0.454
Mixtral 56B Instruct (56B)	0.234	0.326	0.224	0.219	0.219	0.206	0.229	0.207	0.235	0.224
DeepSeek R1 (70B)	0.439	0.437	0.437	0.437	0.436	0.437	0.433	0.432	0.438	0.439

Table 5: Performance comparison of prompting-based approaches across models with varying parameter sizes for Negative categories.

Performance Comparison of LLMs with Cohen’s Kappa (Negative Bias)										
Model	Not Female	Not Muslim Not Hindu	Not Artist	Not Sportsman	Not Vlogger	Not Diplomacy	Not Security Analyst	Not Journalist	Not politician	Not International Relation
Llama 3.2 (3B)	0.055	0.036	0.091	0.098	0.098	0.045	0.152	0.062	0.097	0.098
Qwen 2.5 (7B)	0.688	0.674	0.677	0.681	0.691	0.689	0.758	0.676	0.692	0.682
Mistral (7B)	0.126	0.168	0.108	0.104	0.102	0.082	0.121	0.082	0.131	0.112
Gemini 1.5 Flash (8B)	0.615	0.611	0.630	0.597	0.621	0.608	0.614	0.602	0.586	0.609
Llama 3.1 (8B)	0.738	0.746	0.733	0.737	0.736	0.735	0.706	0.740	0.740	0.744
Llama 3.2 (11B)	0.680	0.733	0.653	0.680	0.704	0.661	0.651	0.594	0.631	0.591
DeepSeek R1 Distill (8B)	0.374	0.474	0.401	0.421	0.400	0.462	0.454	0.445	0.459	0.407
GPT 3.5 Turbo (~20B)	0.160	0.212	0.219	0.186	0.248	0.192	0.217	0.199	0.227	0.231
GPT 4o Mini (~20B)	0.597	0.714	0.641	0.630	0.657	0.668	0.661	0.614	0.634	0.665
Gemini 1.5 Flash (~32B)	0.607	0.612	0.626	0.597	0.611	0.604	0.612	0.603	0.584	0.606
Gemini 2.0 Flash Lite (~40B)	0.591	0.674	0.540	0.564	0.600	0.681	0.712	0.625	0.605	0.579
Llama 3.3 (70B)	0.757	0.769	0.754	0.755	0.774	0.766	0.729	0.745	0.716	0.763
O1 (~200B)	0.632	0.657	0.661	0.670	0.651	0.647	0.624	0.638	0.662	0.616
Gemini 2.0 Flash (~200B)	0.745	0.791	0.659	0.696	0.688	0.777	0.757	0.745	0.710	0.700
Gemini 1.5 Pro (~200B)	0.577	0.645	0.584	0.613	0.615	0.641	0.636	0.612	0.622	0.658
Mixtral 56B Instruct (56B)	0.132	0.179	0.112	0.102	0.103	0.079	0.122	0.081	0.136	0.113
DeepSeek R1 (70B)	0.751	0.742	0.743	0.741	0.737	0.742	0.725	0.722	0.747	0.749

Table 6: Performance comparison of prompting-based approaches across models with varying parameter sizes for Negative categories.

Bias Evaluation for LLMs											
Model	Female vs Not Female	Muslim vs Not Muslim	Hindu vs Not Hindu	Artist vs Not Artist	Sportsman vs Not Sportsman	Vlogger vs Not Vlogger	Diplomacy vs Not Diplomacy	Security Analyst vs Not Security Analyst	Journalist vs Not Journalist	Politician vs Not Politician	International Relation vs Not International Relation
Llama 3.2 (3B)	0.64	1.42	2.01	3.20	4.79	3.07	0.75	3.44	1.90	8.88	4.71
Qwen 2.5 (7B)	0.86	0.85	2.01	2.95	4.49	2.97	1.26	3.70	1.47	7.85	2.41
Mistral (7B)	0.89	1.99	1.39	1.91	4.66	1.39	4.27	1.48	1.39	1.39	3.68
Gemini 1.5 Flash (8B)	11.61	11.89	12.88	12.53	12.79	11.30	12.64	11.40	12.48	13.85	12.91
Llama 3.1 (8B)	2.74	2.53	3.29	2.14	2.62	3.20	2.24	5.00	2.98	5.39	3.04
Llama 3.2 (11B)	5.51	4.96	4.06	2.31	3.27	3.03	2.20	4.65	2.61	7.67	2.55
DeepSeek R1 Distill (8B)	2.25	3.12	2.64	2.13	2.61	2.46	2.43	3.23	2.65	2.87	3.01
GPT 3.5 Turbo (~20B)	21.74	12.35	11.19	14.20	14.09	13.96	31.39	29.78	31.45	30.93	30.97
GPT 4o Mini (~20B)	4.47	5.59	5.74	5.18	5.13	5.73	5.21	5.16	6.76	10.44	4.84
Gemini 1.5 Flash (~32B)	11.62	12.15	12.48	12.09	12.01	12.19	12.18	12.40	11.93	13.97	12.10
Gemini 2.0 Flash Lite (~40B)	12.23	13.70	16.93	12.42	12.28	14.50	13.54	10.61	12.79	14.12	12.68
Llama 3.3 (70B)	2.76	2.77	3.91	3.15	3.37	2.20	2.55	2.43	2.21	9.03	3.69
O1 (~200B)	6.59	5.60	6.61	7.01	6.11	4.79	4.48	5.00	4.95	6.25	5.17
Gemini 2.0 Flash (~200B)	8.83	5.48	4.89	10.57	7.14	8.59	7.31	5.83	7.68	11.04	12.17
Gemini 1.5 Pro (~200B)	21.35	4.08	4.51	5.99	4.40	5.66	4.99	4.96	6.51	7.12	4.89
Mixtral 56B Instruct (56B)	0.80	2.02	1.40	1.82	4.71	1.40	4.22	1.40	1.50	1.20	3.52
DeepSeek R1 (70B)	2.15	2.92	2.05	2.52	1.57	2.17	2.66	2.22	2.42	4.98	2.02

Table 7: Bias evaluation depending on mismatch calculation across models with varying parameter sizes for different categories in different areas.

Performance Comparison of LLMs for reasoning with Cosine Similarity Measure (Positive Bias)											
Model	Female	Muslim	Hindu	Artist	Sportsman	Vlogger	Diplomacy	Security Analyst	Journalist	Politician	International Relation
Llama 3.2 (3B)	0.916	0.546	0.477	0.799	0.819	0.806	0.770	0.810	0.855	0.703	0.820
Qwen 2.5 (7B)	0.977	0.978	0.979	0.977	0.977	0.977	0.977	0.976	0.977	0.978	0.977
Mistral (7B)	0.417	0.418	0.417	0.417	0.417	0.418	0.418	0.418	0.420	0.417	0.417
Gemini 1.5 Flash (8B)	0.921	0.922	0.922	0.921	0.921	0.921	0.921	0.921	0.921	0.922	0.921
Llama 3.1 (8B)	0.979	0.978	0.978	0.979	0.978	0.978	0.977	0.976	0.977	0.977	0.977
Llama 3.2 (11B)	0.984	0.982	0.982	0.983	0.976	0.982	0.982	0.982	0.983	0.982	0.982
DeepSeek R1 Distill (8B)	0.806	0.805	0.806	0.806	0.805	0.805	0.807	0.806	0.806	0.806	0.806
GPT 3.5 Turbo (~20B)	0.973	0.974	0.974	0.973	0.974	0.974	0.973	0.973	0.974	0.974	0.973
GPT 4o Mini (~20B)	0.962	0.961	0.961	0.960	0.963	0.962	0.962	0.962	0.959	0.963	0.960
Gemini 1.5 Flash (~32B)	0.948	0.950	0.948	0.948	0.956	0.949	0.948	0.949	0.945	0.947	0.949
Gemini 2.0 Flash Lite (~40B)	0.555	0.280	0.279	0.550	0.547	0.558	0.283	0.288	0.285	0.286	0.285
Llama 3.3 (70B)	0.996	0.996	0.995	0.996	0.996	0.996	0.995	0.995	0.995	0.995	0.996
O1 (~200B)	0.972	0.971	0.970	0.971	0.971	0.971	0.971	0.971	0.971	0.970	0.971
Gemini 2.0 Flash (~200B)	0.852	0.785	0.662	0.725	0.627	0.701	0.734	0.724	0.766	0.788	0.715
Gemini 1.5 Pro (~200B)	0.968	0.967	0.966	0.967	0.967	0.967	0.968	0.968	0.968	0.967	0.968
Mixtral 56B Instruct (56B)	0.968	0.964	0.957	0.962	0.964	0.960	0.964	0.966	0.964	0.965	0.962
DeepSeek R1 (70B)	0.956	0.943	0.946	0.951	0.948	0.947	0.945	0.944	0.941	0.942	0.941

Table 8: Performance comparison of prompting-based reasoning tasks across models with varying parameter sizes for positive categories.

Performance Comparison of LLMs for reasoning with BERT Score Measure (Positive Bias)											
Model	Female	Muslim	Hindu	Artist	Sportsman	Vlogger	Diplomacy	Security Analyst	Journalist	Politician	International Relation
Llama 3.2 (3B)	0.919	0.803	0.794	0.895	0.907	0.905	0.871	0.903	0.922	0.863	0.913
Qwen 2.5 (7B)	0.944	0.943	0.943	0.943	0.943	0.943	0.943	0.944	0.944	0.943	0.943
Mistral (7B)	0.819	0.819	0.819	0.819	0.819	0.819	0.819	0.819	0.819	0.819	0.819
Gemini 1.5 Flash (8B)	0.925	0.924	0.924	0.923	0.923	0.924	0.924	0.925	0.925	0.924	0.925
Llama 3.1 (8B)	0.951	0.945	0.946	0.947	0.946	0.946	0.945	0.944	0.945	0.944	0.944
Llama 3.2 (11B)	0.954	0.948	0.948	0.951	0.946	0.948	0.948	0.948	0.948	0.948	0.948
DeepSeek R1 Distill (8B)	0.796	0.797	0.798	0.797	0.796	0.797	0.797	0.797	0.797	0.797	0.797
GPT 3.5 Turbo (~20B)	0.941	0.941	0.941	0.941	0.941	0.941	0.941	0.940	0.941	0.941	0.941
GPT 4o Mini (~20B)	0.940	0.940	0.940	0.940	0.940	0.940	0.940	0.940	0.941	0.941	0.940
Gemini 1.5 Flash (~32B)	0.938	0.937	0.937	0.937	0.937	0.936	0.937	0.937	0.936	0.937	0.937
Gemini 2.0 Flash Lite (~40B)	0.922	0.687	0.687	0.920	0.919	0.920	0.688	0.688	0.688	0.688	0.688
Llama 3.3 (70B)	0.989	0.988	0.988	0.988	0.988	0.988	0.988	0.988	0.988	0.988	0.988
O1 (~200B)	0.943	0.943	0.943	0.943	0.943	0.944	0.943	0.943	0.943	0.943	0.943
Gemini 2.0 Flash (~200B)	0.935	0.931	0.927	0.930	0.924	0.926	0.931	0.929	0.932	0.933	0.929
Gemini 1.5 Pro (~200B)	0.933	0.933	0.933	0.933	0.932	0.933	0.934	0.933	0.934	0.933	0.934
Mixtral 56B Instruct (56B)	0.942	0.926	0.925	0.932	0.928	0.925	0.923	0.923	0.923	0.923	0.922
DeepSeek R1 (70B)	0.944	0.937	0.939	0.941	0.940	0.940	0.934	0.939	0.938	0.939	0.938

Table 9: Performance comparison of prompting-based reasoning tasks across models with varying parameter sizes for positive categories.

Performance Comparison of LLMs for reasoning with Cosine Similarity Measure (Negative Bias)										
Model	Not Female	Not Muslim Not Hindu	Not Artist	Not Sportsman	Not Vlogger	Not Diplomacy	Not Security Analyst	Not Journalist	Not Politician	Not International Relation
Llama 3.2 (3B)	0.746	0.580	0.796	0.809	0.832	0.851	0.818	0.863	0.795	0.818
Qwen 2.5 (7B)	0.976	0.980	0.977	0.977	0.976	0.976	0.976	0.977	0.976	0.976
Mistral (7B)	0.418	0.416	0.420	0.419	0.419	0.421	0.419	0.420	0.418	0.419
Gemini 1.5 Flash (8B)	0.921	0.922	0.922	0.922	0.921	0.922	0.921	0.922	0.921	0.922
Llama 3.1 (8B)	0.979	0.977	0.979	0.978	0.979	0.977	0.977	0.977	0.977	0.977
Llama 3.2 (11B)	0.983	0.982	0.983	0.982	0.982	0.982	0.982	0.982	0.982	0.982
DeepSeek R1 Distill (8B)	0.805	0.801	0.805	0.805	0.801	0.806	0.806	0.805	0.806	0.805
GPT 3.5 Turbo (~20B)	0.974	0.974	0.974	0.974	0.973	0.973	0.973	0.974	0.974	0.973
GPT 4o Mini (~20B)	0.959	0.960	0.960	0.961	0.961	0.961	0.963	0.961	0.961	0.962
Gemini 1.5 Flash (~32B)	0.950	0.948	0.950	0.949	0.948	0.947	0.948	0.946	0.949	0.948
Gemini 2.0 Flash Lite (~40B)	0.573	0.272	0.550	0.548	0.551	0.288	0.284	0.288	0.285	0.284
Llama 3.3 (70B)	0.996	0.995	0.996	0.996	0.996	0.995	0.995	0.996	0.995	0.995
O1 (~200B)	0.971	0.970	0.971	0.972	0.971	0.971	0.971	0.971	0.970	0.971
Gemini 2.0 Flash (~200B)	0.772	0.740	0.763	0.715	0.724	0.749	0.766	0.755	0.744	0.732
Gemini 1.5 Pro (~200B)	0.967	0.967	0.967	0.967	0.967	0.967	0.967	0.967	0.967	0.967
Mixtral 56B Instruct (56B)	0.971	0.964	0.962	0.961	0.964	0.962	0.964	0.963	0.964	0.964
DeepSeek R1 (70B)	0.953	0.946	0.948	0.948	0.947	0.945	0.944	0.942	0.941	0.940

Table 10: Performance comparison of prompting-based reasoning tasks across models with varying parameter sizes for Negative categories.

Performance Comparison of LLMs for reasoning with BERT Score Measure (Negative Bias)										
Model	Not Female	Not Muslim Not Hindu	Not Artist	Not Sportsman	Not Vlogger	Not Diplomacy	Not Security Analyst	Not Journalist	Not Politician	Not International Relation
Llama 3.2 (3B)	0.861	0.839	0.900	0.907	0.920	0.929	0.911	0.925	0.910	0.914
Qwen 2.5 (7B)	0.943	0.943	0.943	0.943	0.944	0.943	0.944	0.944	0.943	0.943
Mistral (7B)	0.819	0.819	0.819	0.819	0.819	0.819	0.820	0.819	0.819	0.819
Gemini 1.5 Flash (8B)	0.925	0.924	0.924	0.924	0.925	0.925	0.926	0.926	0.925	0.926
Llama 3.1 (8B)	0.948	0.945	0.947	0.946	0.945	0.944	0.945	0.945	0.944	0.944
Llama 3.2 (11B)	0.951	0.949	0.950	0.949	0.949	0.948	0.948	0.948	0.948	0.948
DeepSeek R1 Distill (8B)	0.797	0.796	0.797	0.796	0.797	0.797	0.797	0.797	0.797	0.797
GPT 3.5 Turbo (~20B)	0.940	0.940	0.941	0.941	0.941	0.941	0.940	0.941	0.941	0.941
GPT 4o Mini (~20B)	0.940	0.940	0.940	0.940	0.940	0.940	0.940	0.940	0.940	0.940
Gemini 1.5 Flash (~32B)	0.937	0.937	0.937	0.937	0.936	0.936	0.937	0.936	0.937	0.937
Gemini 2.0 Flash Lite (~40B)	0.922	0.687	0.922	0.922	0.923	0.688	0.688	0.688	0.687	0.688
Llama 3.3 (70B)	0.989	0.988	0.988	0.988	0.988	0.988	0.988	0.989	0.988	0.988
O1 (~200B)	0.943	0.943	0.943	0.943	0.943	0.943	0.943	0.943	0.943	0.943
Gemini 2.0 Flash (~200B)	0.933	0.930	0.933	0.931	0.931	0.933	0.932	0.932	0.932	0.932
Gemini 1.5 Pro (~200B)	0.933	0.933	0.932	0.933	0.933	0.933	0.933	0.933	0.933	0.933
Mistral 56B Instruct (56B)	0.936	0.924	0.930	0.926	0.925	0.922	0.923	0.923	0.924	0.922
DeepSeek R1 (70B)	0.941	0.938	0.940	0.940	0.940	0.938	0.938	0.937	0.938	0.938

Table 11: Performance comparison of prompting-based reasoning tasks across models with varying parameter sizes for Negative categories.

Models (LLMs)	Prompt 1		Prompt 2		Prompt 3		Prompt 4	
	Female	Not Female	Female	Not Female	Female	Not Female	Female	Not Female
Llama 3.2 (3B)	0.2156	0.2296	0.2189	0.2537	0.2337	0.2476	0.2533	0.2775
Qwen 2.5 (7B)	0.3235	0.2655	0.6091	0.6069	0.5891	0.5546	0.5270	0.5203
Mistral (7B)	0.4170	0.4571	0.4726	0.4388	0.3670	0.4256	0.2067	0.2310
Gemini 1.5 Flash (8B)	0.4202	0.4172	0.4216	0.4189	0.3634	0.3984	0.3647	0.4043
Llama 3.1 (8B)	0.2198	0.2259	0.2346	0.2907	0.2751	0.2470	0.2721	0.2323
Llama 3.2 (11B)	0.4012	0.4667	0.4049	0.4540	0.3154	0.6048	0.4768	0.4989
DeepSeek R1 (8B)	0.3013	0.3410	0.3434	0.3513	0.3244	0.2906	0.2855	0.3168
GPT 3.5 Turbo (~20B)	0.2796	0.2675	0.2906	0.2328	0.2014	0.2078	0.1996	0.2034
GPT 4o Mini (~20B)	0.4512	0.4494	0.4164	0.4235	0.3789	0.3791	0.2493	0.2499
Gemini 1.5 Flash (~32B)	0.4338	0.4315	0.4337	0.4586	0.2958	0.3029	0.2924	0.2921
Gemini 2.0 Flash Lite (~40B)	0.4519	0.4328	0.4412	0.4558	0.3047	0.3336	0.3361	0.3456
Llama 3.3 (70B)	0.4549	0.4498	0.4391	0.4609	0.4362	0.6051	0.4514	0.4525
O1 (~200B)	0.4208	0.4580	0.4597	0.4544	0.4608	0.4621	0.4574	0.4609
Gemini 1.5 Pro (~200B)	0.4338	0.4330	0.4230	0.4276	0.3886	0.3887	0.3643	0.3601
Gemini 2.0 Flash (~200B)	0.4514	0.4514	0.4049	0.4385	0.3627	0.3650	0.2609	0.2495
Mistral 56B Instruct (56B)	0.3807	0.3831	0.6544	0.6831	0.3049	0.3532	0.2530	0.2388
DeepSeek R1 (72B)	0.6206	0.6255	0.7555	0.7921	0.6333	0.5666	0.5310	0.5156

Table 12: Performance and Bias Evaluation of 17 Large Language Models (LLMs) on Gender key area Using F1 Scores Across Four Prompt Levels (level-1 to level-5) in the TELeR Taxonomy. The models were assessed separately for “Female” and “Not Female” categories to capture potential bias and reasoning depth in gender-related prompts.

Models (LLMs)	Prompt 1		Prompt 2		Prompt 3		Prompt 4	
	Female	Not Female	Female	Not Female	Female	Not Female	Female	Not Female
Llama 3.2 (3B)	0.2450	0.2468	0.2827	0.2978	0.2623	0.2901	0.2825	0.2732
Qwen 2.5 (7B)	0.1997	0.1984	0.8185	0.8140	0.7035	0.6710	0.6844	0.5932
Mistral (7B)	0.2539	0.2830	0.2483	0.2995	0.3533	0.3209	0.2897	0.3348
Gemini 1.5 Flash (8B)	0.5954	0.5662	0.6084	0.6032	0.3985	0.4974	0.4157	0.5411
Llama 3.1 (8B)	0.2325	0.2493	0.3174	0.3698	0.3460	0.3587	0.3727	0.3883
Llama 3.2 (11B)	0.2811	0.4261	0.6101	0.7080	0.1440	0.2780	0.2687	0.3417
DeepSeek R1 (8B)	0.2597	0.2707	0.3785	0.3787	0.3456	0.3023	0.3136	0.2743
GPT 3.5 Turbo (~20B)	0.2521	0.2354	0.2432	0.1668	0.1783	0.1857	0.1864	0.1929
GPT 4o Mini (~20B)	0.7510	0.7528	0.5985	0.6221	0.4932	0.4921	0.4318	0.4378
Gemini 1.5 Flash (~32B)	0.6883	0.6823	0.7208	0.7633	0.2974	0.3138	0.2974	0.2977
Gemini 2.0 Flash Lite (~40B)	0.7901	0.7184	0.7503	0.6567	0.2913	0.3560	0.3893	0.4044
Llama 3.3 (70B)	0.8047	0.7841	0.7438	0.8309	0.7319	0.8045	0.7852	0.7906
O1 (~200B)	0.8397	0.8257	0.8293	0.8078	0.8327	0.8395	0.8243	0.8367
Gemini 1.5 Pro (~200B)	0.6208	0.6404	0.5928	0.5832	0.5179	0.5093	0.4601	0.4554
Gemini 2.0 Flash (~200B)	0.7841	0.7841	0.5651	0.6475	0.4247	0.4348	0.2878	0.2972
Mistral 56B Instruct (56B)	0.3378	0.2600	0.3247	0.2215	0.1242	0.1908	0.1245	0.1882
DeepSeek R1 (72B)	0.8524	0.8731	0.7123	0.7849	0.5961	0.6117	0.6062	0.5758

Table 13: Performance and Bias Evaluation of 17 Large Language Models (LLMs) on Gender key area Using Cohen’s Kappa Scores across Four Prompt Levels (level-1 to level-5) in the TELeR Taxonomy. The models were assessed separately for “Female” and “Not Female” categories to capture potential bias and reasoning depth in gender-related prompts.

Models (LLMs)	Prompt 1	Prompt 2	Prompt 3	Prompt 4
	Female vs Not Female	Female vs Not Female	Female vs Not Female	Female vs Not Female
Llama 3.2 (3B)	16.779	14.66	12.35	11.53
Qwen 2.5 (7B)	9.259	1.201	8.66	11.61
Mistral (7B)	17.17	15.44	30.86	14.56
Gemini 1.5 Flash (8B)	13.01	11.79	21.43	20.61
Llama 3.1 (8B)	22.78	13.22	9.032	9.811
Llama 3.2 (11B)	6.643	6.708	6.481	7.14
DeepSeek R1 (8B)	14.28	11.30	9.282	12.23
GPT 3.5 Turbo (~20B)	15.70	22.29	9.226	8.684
GPT 4o Mini (~20B)	4.044	4.782	4.79	5.730
Gemini 1.5 Flash (~32B)	8.759	3.527	5.935	6.803
Gemini 2.0 Flash Lite (~40B)	10.57	6.714	14.04	13.97
Llama 3.3 (70B)	0.697	0.712	1.413	2.097
O1 (~200B)	4.967	1.195	0.941	1.560
Gemini 1.5 Pro (~200B)	9.606	4.193	7.649	9.404
Gemini 2.0 Flash (~200B)	5.219	6.056	11.35	11.14
Mistral 56B Instruct (56B)	26.92	21.92	19.75	22.85
DeepSeek R1 (72B)	1.954	0.826	2.931	4.436

Table 14: Bias Evaluation of 17 Large Language Models (LLMs) between two categories of Gender key area using Mismatch calculation across Four Prompt Levels (level-1 to level-5) in the TELeR Taxonomy. The models were assessed for “Female” vs “Not Female” to capture potential bias and reasoning depth in gender-related prompts.

Model Name	Number of Labels	Category	F1 Score	Cohen's Kappa	Mismatch (%)
DeepSeek R1 8B	5	Politician	68.16	56.35	17.05
	5	Not Politician	67.12	54.79	
	4	Politician	78.52	78.30	14.24
	4	Not Politician	75.57	73.12	
	3	Politician	92.17	89.64	13.13
	3	Not Politician	80.01	73.23	
	2	Politician	98.83	93.33	7.45
	2	Not Politician	97.33	95.38	
Gemini 1.5 Flash 8B	5	Politician	94.22	83.43	2.97
	5	Not Politician	94.97	86.36	
	4	Politician	93.91	82.83	2.92
	4	Not Politician	93.41	83.86	
	3	Politician	96.86	90.17	2.04
	3	Not Politician	97.05	91.39	
	2	Politician	98.74	97.15	0.36
	2	Not Politician	98.41	96.72	
GPT-3.5 Turbo	5	Politician	91.06	75.06	6.40
	5	Not Politician	92.47	78.16	
	4	Politician	93.49	78.97	4.82
	4	Not Politician	92.86	77.94	
	3	Politician	95.57	84.48	3.48
	3	Not Politician	95.23	84.83	
	2	Politician	96.90	90.96	1.44
	2	Not Politician	97.59	93.01	
LLaMA 3.2 3B	5	Politician	90.46	75.38	5.12
	5	Not Politician	90.05	75.17	
	4	Politician	86.76	64.78	4.44
	4	Not Politician	86.44	66.03	
	3	Politician	94.48	84.20	4.20
	3	Not Politician	96.09	88.85	
	2	Politician	96.63	90.08	2.76
	2	Not Politician	97.79	93.53	
Mistral 7B	5	Politician	88.47	69.03	4.70
	5	Not Politician	89.97	71.62	
	4	Politician	89.74	71.26	1.54
	4	Not Politician	89.62	70.95	
	3	Politician	92.10	75.71	3.90
	3	Not Politician	94.69	79.29	
	2	Politician	95.04	84.56	2.72
	2	Not Politician	94.28	79.99	
Qwen 2.5 7B	5	Politician	94.75	84.11	1.30
	5	Not Politician	94.88	84.80	
	4	Politician	94.91	83.98	0.94
	4	Not Politician	94.74	83.93	
	3	Politician	95.70	86.25	0.68
	3	Not Politician	95.69	86.61	
	2	Politician	96.73	90.17	0.80
	2	Not Politician	96.43	89.04	

Table 15: Performance of various LLMs on the Politician/Not Politician classification task across different label configurations.